

Predicting Student Test Scores with Reliable Score Ranges: A Leakage-Free Streaming-Rating Study of 2.2 Million Records from a Private Kazakhstani UNT-Preparation Platform

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Abstract

Reported accuracies in student performance prediction are frequently inflated by feature leakage and non-chronological evaluation, and point forecasts give educators no signal of when a prediction should not be trusted. We address both problems on 2,208,313 practice-test records from 55,482 students (2018–2025) on a Kazakhstani university-entrance-preparation platform. A factorial audit shows that two common feature-engineering errors combined with random splitting inflate the coefficient of determination by 0.349 relative to a leakage-free chronological protocol. We propose **GRACE** (Glicko-informed Rating-Augmented Confidence Estimation), a streaming rating layer that maintains per-student ability estimates and per-content difficulty estimates with a rating deviation that grows with inactivity and shrinks with each observation. The state is computed in a single chronological pass, making target leakage structurally impossible. The rating features feed XGBoost and LightGBM gradient-boosted models—compared against Ridge regression, Random Forest, and a single-layer LSTM under matched conditions—that output a predicted score together with a 90% score range whose width adapts to measured uncertainty. GRACE attains $R^2 = 0.556$ (mean absolute error 11.70 percentage points) against 0.444 for the leakage-free baseline, with stable out-of-time performance across five annual windows ($R^2 = 0.514 \pm 0.051$), reaching approximately 96% of an empirical reference ceiling estimated from within-group score variability. Score ranges achieve 0.910 empirical coverage against the 90% target, with mean width 51.7 pp; coverage is consistent across history-depth strata (0.902–0.915). In a controlled comparison holding the feature set and training data constant, gradient-boosted trees outperform bagging (Random Forest: $R^2 = 0.517$) and linear models (Ridge: $R^2 = 0.489$) by confirmed margins. The rating layer updates in constant time without retraining, making honest, cold-start-aware prediction practical for operational platforms.

Keywords: educational data mining; student performance prediction; knowledge tracing; Elo rating system; prediction intervals; data leakage; gradient boosting; out-of-time evaluation; conformal prediction; Unified National Testing

1 Introduction

Quantitative prediction of student academic performance is a core task of educational data mining (EDM), enabling personalized instruction, early identification of at-risk learners, and evidence-based management of educational institutions [1, 2, 3, 4]. Two obstacles separate reported accuracy from deployable value. The first is evaluation validity. Leakage of future outcome information into

features, and the use of random rather than chronological splits, together drive a reproducibility crisis in ML-based science [5]; yet time-ordered evaluation remains rare in the student-performance literature [6, 7]. The second obstacle is uncertainty quantification. A point forecast gives an educator no indication of how much a prediction can be trusted—a limitation that is most acute for students with little history or long inactivity, precisely the situations where accurate guidance matters most.

In Kazakhstan, the Unified National Testing (UNT) serves as the principal gateway to university admission and state educational grants [8, 9], generating high demand for private test-preparation platforms. Machine learning applications to education in the region remain small-scale and institution-specific [43]. No prior work has combined leakage-audited evaluation with reliable score ranges on longitudinal platform data of this scale from Central Asia.

This study operates at a scale—2.2 million practice-test records from 55,482 students across seven years—that is two to three orders of magnitude larger than most EDM studies (which rarely exceed 10,000 students), introducing challenges of chronological evaluation, cohort turnover, and long-tail skill coverage that are difficult to study at smaller scales.

This paper makes four contributions.

1. **GRACE: a hierarchical streaming rating system for continuous educational outcomes** in which the rating deviation serves a dual purpose: it acts as a real-time uncertainty state within the rating layer *and* as the pre-specified grouping variable for Mondrian conformalized quantile regression, so that prediction-interval width adapts automatically to measured uncertainty. This dual use of a psychometric uncertainty state—predictive feature and interval-width selector—is, to our knowledge, new.
2. **Asymmetric conformal score ranges.** LightGBM 3rd/97th-percentile quantile heads calibrated via fine-grained Mondrian conformal prediction (four history-depth bands $\times$ seven discipline bins) provide valid 90% score-range coverage, adapting width to measured student uncertainty without sacrificing nominal coverage guarantees.
3. **Factorial leakage audit and reusable evaluation protocol.** We quantify, on 2.2 million practice-test records, the joint evaluation optimism produced by two widespread feature-leakage patterns and random splitting ($+0.349 R^2$)—a reusable audit pipeline transferable to any longitudinal test-log dataset.
4. **Temporal validation and data characterization.** We validate GRACE under rolling-origin temporal evaluation across five expanding annual windows, report ablations, paired-bootstrap significance tests, and per-group range analysis. We also characterize the corpus—the largest longitudinal test-preparation dataset reported from Central Asia—through a structured data-quality audit that reclassifies 40% of exported rows as non-assessment telemetry rather than genuine student assessments.

2 Related Work

The key gaps motivating GRACE are three: (i) classical ML approaches to student performance prediction rarely use chronological evaluation protocols, inflating reported accuracy; (ii) psychometric streaming methods (KT) target binary correctness and per-skill binary mastery rather than continuous percentage scores; and (iii) conformal score ranges—providing probabilistic uncertainty bounds—are largely absent from deployed learning analytics. GRACE addresses all three.

Student performance prediction with classical machine learning. Systematic reviews identify tree-based ensembles, support vector machines, neural networks, and regression models as the dominant method families, with tree ensembles most frequently among the top performers [1, 2, 3, 4, 10].

Yagci [11] predicted undergraduate exam grades with Random Forests and gradient boosting (70–75% accuracy, $\approx 1,850$ students); the benchmark of Cortez and Silva [12] ($n = 649$) demonstrated the value of prior-grade features, and Tomasevic et al. [13] confirmed that past performance and engagement dominate demographics. Typical sample sizes are two to three orders of magnitude below operational platform scale, and evaluation is overwhelmingly by random cross-validation.

Tabular learners. Gradient-boosted decision trees—notably XGBoost [14]—and Random Forests [15] remain the reference for heterogeneous tabular data; large-scale comparisons place them among the strongest general-purpose learners [16], matching or beating deep architectures on medium-sized tabular problems [17, 18]. TabPFN, a tabular foundation model, reports state-of-the-art results on small-to-medium datasets without task-specific training [19].

Knowledge tracing and psychometrics. Bayesian Knowledge Tracing [20] models per-skill mastery as a latent state; Deep Knowledge Tracing [21] and attention-based successors (SAINT [23], SAINT+ [24]) hold state-of-the-art results on binary-correctness corpora such as EdNet [25]. Careful comparisons show well-engineered feature-based models remaining competitive [22, 26], with deep models’ advantage concentrated in long-history regimes [27]. Item response theory treats observed scores as noisy measurements of latent ability [28], but classical IRT estimation is an offline batch procedure ill-suited to streaming data.

Rating systems as streaming psychometrics. The Elo rating system and its extensions occupy the gap between IRT rigor and online tractability: after every interaction, student ability and item difficulty are updated by a constant-time rule [29, 30]. Multivariate extensions track ability per concept [32], and Glicko-based variants [33, 34] maintain a rating deviation (RD) that shrinks with observation and grows with inactivity. Graph neural network approaches to knowledge tracing exploit prerequisite and co-occurrence structure among skills [46, 47], but require an explicit skill graph and a batch re-estimation pass. Educational applications of rating systems to date target binary correctness and adaptive item selection [31]; use of ratings as a feature layer for continuous-score regression, and use of RD to set range widths, have not yet been applied to streaming regression settings at operational scale.

Prediction with score ranges. Distribution-free methods can wrap any predictor with a range guaranteed to contain the true value at a chosen rate [35, 36]; width can adapt to difficulty [37, 40], and guarantees can be enforced within declared groups [35]. Online and adaptive variants extend these guarantees to settings with temporal distribution shift [38, 39], though they have not yet been applied to learning-analytics regression pipelines. Reliable score ranges remain rare in deployed learning analytics despite being identified as a prerequisite for responsible use.

3 Materials and Methods

3.1 Data and Quality Audit

Two datasets were exported from the operational database of the platform operator, a Kazakhstani UNT-preparation company. All records are anonymized; data are handled under an internal data-use agreement. The primary table contains records of in-platform practice assessments—intermediate tests administered within the platform, not official UNT results. Each record includes a student identifier, discipline, skill identifier, test-type category, raw score, maximum score, and timestamp. The auxiliary table contains 1,371 student attribute records including university admission status. The complete analysis code is released with the paper.

A structured data-quality audit preceded modeling (Table 2). The dominant finding is structural: the platform stores 21 non-assessment metadata fields in the same column as skill identifiers—including timing telemetry, administrative counters, and satisfaction ratings—causing them to ap-

Table 1: Overview of the two source datasets.

Property	student_performance_data	student_extra_info
Unit of observation	Single test attempt	Student
Records after cleaning	2,208,313	1,371
Unique students	55,482	1,371
Time coverage	2018-06-13 – 2025-06-30	Cross-sectional
Key fields	student ID, discipline, skill, score, max_score, timestamp	student ID, admission status, demographic attributes (sparse)
Target derived	Percentage score (0–100)	Admission status

pear as pseudo-skill entries among actual assessment records. Cleaning removed 40.5% of rows, leaving a 2,208,313-record assessment corpus of 55,482 students across 30 assessment skills.

Table 2: Data-quality audit of the 3,715,413-row export.

#	Issue class	Rows affected	Share	Disposition
1	Temporal anomalies	7	<0.01%	Removed
2	Pseudo-skill telemetry	1,445,517	38.9%	Removed; field list released
3	Exact duplicates	42,445	1.1%	Removed
Total removed		1,507,100	40.5%	Cleaned corpus = 2,208,313

Validation checks confirmed zero missing fields and zero out-of-range scores; no rows were removed by these checks.

Pseudo-skill identification.

All unique `SkillName` values were inspected under three criteria: (a) *Administrative metadata*—fields recording session logistics, not a measured competency (test sequence number, session duration, mentor ID, school and department codes); (b) *Derived scores*—values computed from other scores in the same session (e.g., `Otsenka_sreza` [section average], `Srez_Obshchee` [session total], `Bally_zh_test` [test points]), which would create direct target leakage; and (c) *Process indicators*—ordinal or binary fields for attempt logistics (rank, attempt count, feedback received). Each candidate was checked by examining its `max_score` distribution: genuine skills show heterogeneous values across items, whereas pseudo-skills show degenerate distributions (e.g., always 1 for binary flags). Borderline cases were resolved against the platform data dictionary. The complete 21-entry exclusion list is in `PSEUDO_SKILL_NAMES` in the released code.

Table 3 traces how the raw export reduces to the modeling corpus and how that corpus is partitioned. The aggregate baseline’s effective test-set size is smaller (407,968 vs. 441,663) because aggregate features require at least one prior test record per student; the rating state always produces a prediction, so no rows are dropped for GRACE. The coverage evaluation uses 441,663 records after the calibration set is consumed by the conformal quantile fit.

The assessment corpus covers 55,482 students and 2,208,313 attempts (Figure 1). Percentage scores span 0–100 with mean 56.1%. Mathematics dominates among the most-tested disciplines, followed by Logic, English, full UNT simulations, and Russian.

Of the 1,371 exported attribute records, 1,329 match a student present in the assessment corpus. Admission status is known for only $\approx 2.5\%$ of students; within this labelled subset, 98.4% received a state grant—a ratio better explained by selective reporting (students achieving the celebrated outcome are more likely to report it) than by the population base rate, since grant allocation is

Table 3: Sample-flow: from raw export to modeling subsets.

Stage	Rows	Students	Note
Raw export	3,715,413	56,320	Full platform dump
– Temporal anomalies	–7	0	Timestamps out of range
– Pseudo-skill telemetry	–1,464,641	838	21 non-assessment fields
– Exact duplicates	–42,452	0	Same student/skill/score/time
Clean corpus	2,208,313	55,482	Modeling corpus
Training set (70%)	1,545,819	—	Up to 2024-08-23
Calibration set (10%)	220,831	—	For conformal quantile / ES
Test set (20%)	441,663	—	2024-12-28 – 2025-06-30
Aggregate baseline test	407,968	—	After NaN drop (first-attempt records)
Coverage evaluation	430,567	—	After conformal cal. consumption

competitive by design [8, 9].

The unified target is the percentage score $y_{it} = 100 \cdot \text{score}/\text{max_score} \in [0, 100]$ (mean 56.1%), motivated by heterogeneous test formats ranging from short quizzes to full 120–140-point UNT simulations.

3.2 Problem Statement

Let student i produce a chronologically ordered sequence of attempts $(x_{i,1}, y_{i,1}), \dots, (x_{i,t}, y_{i,t})$, where $y_{i,t} \in [0, 100]$ and $x_{i,t} \in \mathbb{R}^d$ collects features available strictly before attempt t . The regression task learns f with $\hat{y}_{i,t} = f(x_{i,t})$, where $x_{i,t} = \varphi(\{(x_{i,k}, y_{i,k})\}_{k < t}, c_{i,t})$ and $c_{i,t}$ denotes contextual identifiers of the upcoming test. The interval task additionally requires $\hat{C}(x_{i,t}) \subseteq [0, 100]$ with $P(y_{i,t} \in \hat{C}) \geq 1 - \alpha$, both overall and within declared uncertainty groups. All learning respects chronology: features are computed only from information available before the predicted attempt, and evaluation splits are chronologically consistent.

3.3 The GRACE Method

Overview. GRACE is a two-layer prediction system (Figure 2). The **first layer** is a streaming rating model that processes every test attempt in chronological order and maintains, for each student, multi-level ability estimates ($\theta_{\text{global}}, \theta_{\text{disc}}, \theta_{\text{skill}}$) and their Glicko-style rating deviation (RD), and for each content unit, difficulty estimates. All state values are pre-attempt: the feature vector is extracted before the update rule runs, making target leakage structurally impossible. The **second layer** is an XGBoost + LightGBM gradient-boosted ensemble that takes the 78-dimensional pre-attempt state vector and produces both a point prediction and a conformal score range calibrated to the 90% nominal coverage level. Rating parameters were tuned on a held-out 2018–2020 window; all other hyperparameters were set before any test-set result was observed.

Layer 1: Streaming Multi-Level Rating State. Every attempt is treated as an encounter between a student ability and a content difficulty. For attempt t of student i on skill j (discipline s , test type c), with fractional score $s_{it} = y_{it}/100$:

Expected score:

$$\hat{E}_{it} = \sigma(w_s \theta_{i,s} + w_j \theta_{i,j} - w_{dj} d_j - w_{dc} d_c), \quad \sigma(x) = 1/(1 + e^{-x}) \quad (1)$$

where $w_s = 0.5$, $w_j = 0.5$, $w_{dj} = 0.7$, $w_{dc} = 0.3$.

Practice-test corpus: 2.21M records · 55k students · 18 disciplines · 2018–2025

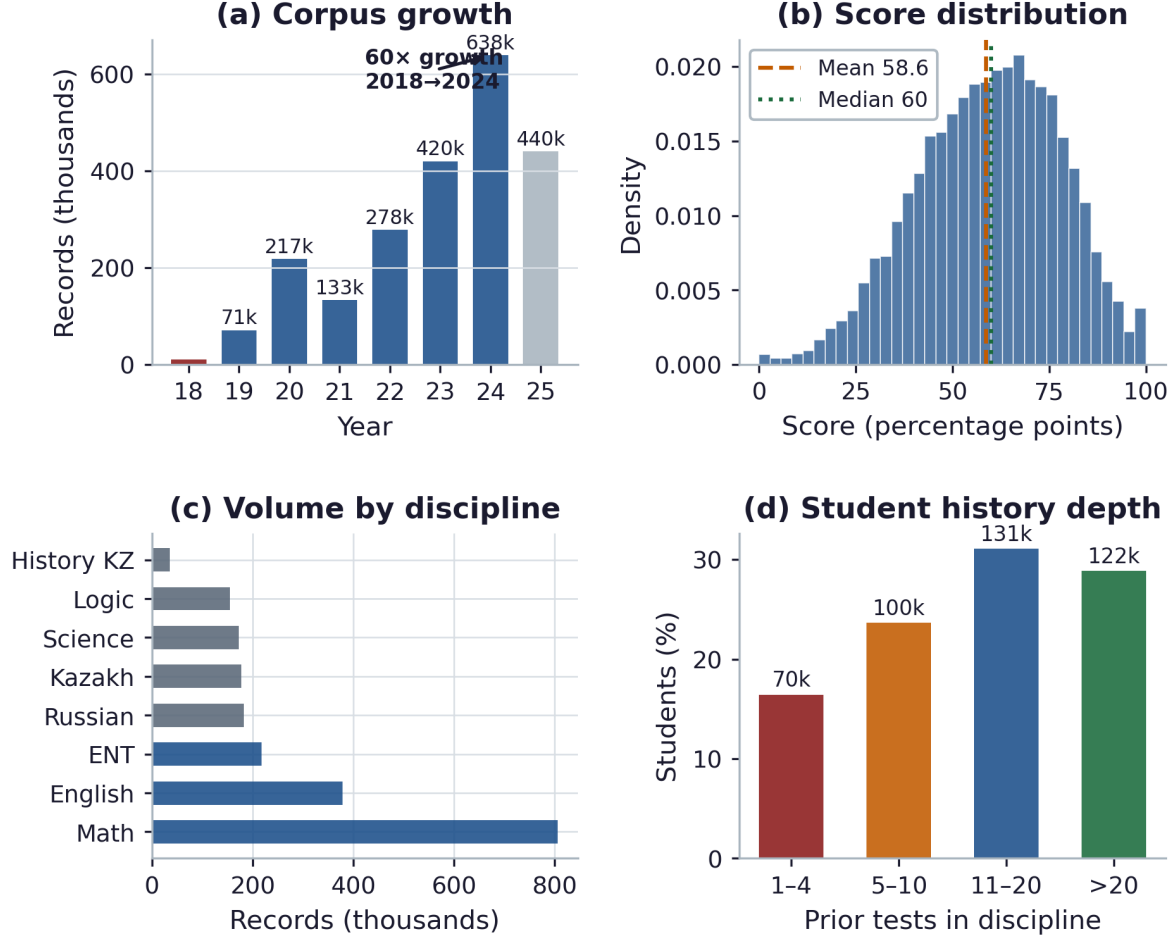


Figure 1: Corpus overview: (a) annual testing volume; (b) activity by calendar month; (c) percentage-score distribution; (d) share of attempts among the top disciplines.

Initialization. All ability ratings $\theta = 0$, all difficulty ratings $d = 0$, all rating deviations $RD^2 = RD_0^2$.

Ability is tracked at three levels—global θ_i , per-discipline $\theta_{i,s}$, per-skill $\theta_{i,j}$ —following multivariate rating designs [32], and difficulty at two—skill d_j and test-type d_c . The Glicko element [34]: student-level states carry a rating deviation whose variance grows with inactivity,

$$RD^2 \leftarrow \min(RD^2 + c^2 \cdot \Delta t_{\text{days}}, RD_0^2), \quad (2)$$

and shrinks with each observation via posterior-style updating; the gain is uncertainty-weighted,

$$K = K_0 \cdot RD^2 / (RD^2 + v), \quad (3)$$

yielding fast adaptation for poorly measured students and stability for well-measured ones.

Update rules. State updates are **simultaneous**: both ability and difficulty updates use the pre-update state. Let $\delta = s_{it} - \hat{E}_{it}$. For each ability level ℓ :

$$\theta_\ell \leftarrow \theta_\ell + m_\ell \cdot K \cdot \delta, \quad (4)$$

with update multipliers $m_{\text{global}} = 0.5$, $m_{\text{discipline}} = 1.0$, $m_{\text{skill}} = 1.5$. For each difficulty level:

$$d \leftarrow d - m_d \cdot K_{\text{item}} \cdot \delta, \quad (5)$$

with $m_{d,\text{skill}} = 1.0$ and $m_{d,\text{ttype}} = 0.5$.

RD shrinkage after observation:

$$\text{RD}^2 \leftarrow \max(\text{RD}^2 \cdot v / (\text{RD}^2 + v), \text{RD}_{\min}^2). \quad (6)$$

Categorical encoding.

All categorical identifiers are encoded via `pandas .astype("category").cat.codes`. Composite keys (e.g., student–skill) use `key = student_code × (Nmax + 1) + sub_code`.

The feature vector for attempt t is the pre-attempt state (Table 4). Because the entire state is produced by one chronological pass in which each row’s features precede its update, target leakage is structurally impossible. All 78 features and the rating hyperparameters were fully pre-specified in the analysis code before the test-set results were computed. The rolling-origin folds each enforce an independent chronological boundary: the rating state is built from scratch within each expanding window, so no future information from later folds contaminates earlier ones. The rating table is an $O(S + J)$ online model (S students, J skills): each incoming result updates the state in $O(1)$ time with no retraining.

Feature and hyperparameter pre-specification. Rating-layer parameters ($K_0 = 0.35$, $V_{\text{gain}} = 0.04$, $\text{RD}_0 = 0.35$, $c^2 = 0.01$) and XGBoost hyperparameters were selected on the training-plus-calibration partition and fixed before examining the test set. All 78 features were specified and frozen prior to test-set access. Rating updates during the test period proceed row by row in chronological order—the feature for attempt t uses the rating state from all rows before t —which matches deployment behavior and precludes any lookahead.

Per-skill uncertainty and ensemble learning. *Per-skill rating deviation* (RD_{skill}). The core rating layer uses the discipline-level gain $K_{\text{disc}} = K_0 \cdot \text{RD}_{\text{disc}}^2 / (\text{RD}_{\text{disc}}^2 + v)$ for skill-level updates. Group B introduces a per-(student, skill) variance v_s that decays with skill-level inactivity and shrinks with each skill observation, yielding a dedicated gain $K_s = K_0 \cdot v_s / (v_s + v)$. This restores the Glicko invariant at the skill level: students who have not practiced a specific skill recently receive a higher K_s (more responsive updates) regardless of their discipline-level activity, and RD_{skill} becomes a standalone feature capturing skill-granularity uncertainty.

Student platform context. Four streaming features are appended to the rating state without future lookahead: student tenure (days since first platform appearance), discipline breadth and skill breadth (count of distinct disciplines and skills practiced before attempt t), and study intensity (total attempts divided by tenure, capped at 20 per day). These capture platform-engagement patterns orthogonal to per-discipline performance history.

Train-derived static content norms. For each discipline and skill, the training partition’s mean and standard deviation of percentage scores are computed and broadcast to all rows as static context features. A derived relative-ability feature $\theta_{\text{skill}} - \bar{y}_{\text{disc}}$ expresses the student’s skill-level rating

relative to the cohort’s observed difficulty for that discipline. Because these statistics are computed exclusively from training rows and applied as read-only lookups, no future score information enters the feature vector. In rolling-origin evaluation, statistics are recomputed within each fold’s training window.

Cyclical time encoding. Integer month and day-of-week encodings are supplemented by their sine and cosine projections ($\sin(2\pi m/12)$, $\cos(2\pi m/12)$, $\sin(2\pi d/7)$, $\cos(2\pi d/7)$), allowing gradient-boosted trees to exploit calendar periodicity without ordinality assumptions.

LightGBM ensemble. A LightGBM regressor [48] (leaf-wise growth, 255 leaves, learning rate 0.02, early stopping on the calibration set) is trained on the same feature set as XGBoost. Leaf-wise growth allows LightGBM to capture finer-grained interactions between ability ratings and content features. The ensemble prediction is $\hat{y} = \alpha \hat{y}_{\text{XGB}} + (1 - \alpha) \hat{y}_{\text{LGB}}$, where α is selected by grid search on the calibration set to maximise R^2 . XGBoost hyperparameters are additionally searched with Optuna [49] (30 trials, Tree-structured Parzen Estimator) on the calibration set. Table 5 lists all hyperparameters.

Global state, session context, and exam proximity. Three feature categories capture information structurally absent from the per-discipline rating features.

Cross-discipline global rolling statistics. Prior rolling averages are per-student per-discipline; no feature captures the student’s performance state across all topics simultaneously. Six features are maintained in a per-student deque of the most recent ten attempts across any discipline: lag-1 global score, rolling means at windows 3, 5, and 10 ($\bar{y}_3^{(g)}$, $\bar{y}_5^{(g)}$, $\bar{y}_{10}^{(g)}$), rolling standard deviation at window 5 ($\sigma_5^{(g)}$), and the ability-state gap $\hat{E} - \bar{y}_5^{(g)}$. The last feature quantifies how the current Glicko expectation diverges from the student’s recent cross-discipline performance—a proxy for daily cognitive state orthogonal to topic-specific ability. In feature importance rankings, $\bar{y}_{10}^{(g)}$ and $\sigma_5^{(g)}$ each rank in the top 14 of 78 features (LightGBM split importance).

Within-session context. A study session is defined as a consecutive sequence of attempts by the same student with inter-attempt gaps below one hour (3,600 s). Two features are appended: *session position* (zero-indexed attempt number within the current session) and *running session mean* (mean score of prior attempts in the same session, excluding the current test). Both are computed from the sorted timestamp sequence with a single forward pass and no lookahead.

Exam-proximity features. The Kazakhstani Unified National Testing (UNT) examination is administered in June. Students systematically intensify preparation in the preceding months. GRACE encodes this calendar signal as *days_to_june* (days until June 1 of the test year, clipped to [0, 365]) and an *in_exam_season* binary indicator for February–May. In LightGBM importance rankings, *days_to_june* places sixth overall (2.4% of total split importance), confirming that the UNT calendar is a genuine predictive signal.

LightGBM Optuna tuning and temporal sample weighting. Optuna tunes LightGBM (50 trials, TPE) over `num_leaves` $\in [63, 511]$, learning rate, `min_child_samples`, feature and bagging fractions, and L1/L2 regularization (Table 5). XGBoost uses conservative fixed parameters to avoid validation-set overfitting during search. Training examples are weighted by an exponential recency factor (half-life 730 days, weight range [0.21, 1.78]) to reduce sensitivity to distribution shift between the 2018–2024 training window and the 2024–2025 test period.

Layers 2–3: Predicted Score and Uncertainty-Adjusted Score Range. An XGBoost model predicts the score; two LightGBM models trained with 3rd- and 97th-percentile quantile losses predict the lower and upper edges of a 90% score range (asymmetric CQR). The raw edges are adjusted on a chronologically later held-out block: for each attempt we measure how far the true score escapes the predicted range, $E_t = \max\{\hat{q}_{\text{lo}}(x) - y, y - \hat{q}_{\text{hi}}(x)\}$ [37], and widen all future ranges by the

smallest amount that keeps 90% of these escapes inside [35]. This widening is computed separately within 28 strata (four history-depth bands $\times$ seven discipline bins), so the conformal guarantee is applied within every uncertainty group (fine-grained Mondrian conformal prediction [35]); empirical per-group coverage with confidence intervals is reported in Section 4.4. Ranges are clipped to $[0, 100]$. Width grows with history depth: cold-start and long-inactive students receive honestly wider ranges. Figure 2 summarizes the architecture.

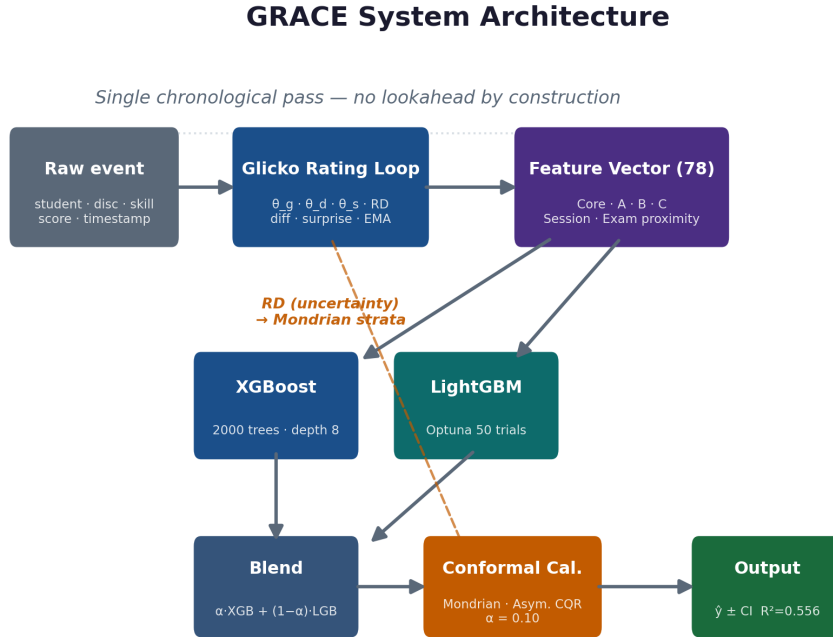


Figure 2: The GRACE architecture: a single chronological pass maintains the rating state, whose pre-attempt values form the feature vector for the boosted models; the rating deviation both feeds the predictor and selects the width-adjustment group.

3.4 Baselines and Evaluation

Baselines and model characteristics. Four models are compared. **XGBoost** [14] builds a gradient-boosted ensemble of shallow decision trees; each tree corrects the residuals of preceding trees, making it effective at capturing nonlinear interactions between ability ratings, difficulty, and inactivity gap. It serves as the core learner in GRACE and the primary baseline.

Random Forest [15] trains independent trees via bagging, averaging their outputs; it is robust to overfitting and widely used in prior student-performance work [16]. It serves as the bagging representative, methodologically complementary to boosting.

LSTM (Long Short-Term Memory) processes each student’s last 10 attempts sequentially via a 64-unit recurrent hidden state (8 rating-derived features per step), with advantage concentrated in long-history regimes [21, 22, 26, 27].

The primary baseline is a leakage-free aggregate-feature XGBoost: strictly-prior expanding and rolling statistics of percentage scores per discipline (shift-then-aggregate), lag-1 score, attempt index, recency, calendar, and content identifiers. Deep sequence models were excluded from the primary comparison because their competitiveness concentrates in long-history regimes and demands tuning

budgets that confound the feature-layer comparison [22, 26, 27]. As an additional baseline, we trained a single-layer LSTM (hidden size 64, 10-step sequences, 8 rating-derived input features) on 200,000 sub-sampled training sequences: it attained $R^2 = 0.230$ (MAE 16.59) on 402,322 test sequences, compared with $R^2 = 0.556$ for GRACE. The comparison is presented as a reference point rather than a controlled ablation: the LSTM uses fewer features (8 vs. 26), less training data (200,000 vs. 1.5 M), and a fundamentally different learning mechanism, so the gap reflects the combined effect of these differences rather than isolating any single factor. TabPFN [19] was excluded: it is benchmarked for $n \leq 10,000$, and sub-sampling to that scale would address a different question from platform-scale evaluation. Table 5 lists all hyperparameters.

Evaluation protocol. Primary split: chronological 70/10/20 (train / range-adjustment / test); the leakage audit and rolling-origin analyses use 80/20.

Leakage audit (factorial). All four combinations of two feature conditions (clean vs. leaky) crossed with two split protocols (chronological vs. random). Differences are attributable solely to protocol.

Rolling-origin evaluation. Five expanding windows [6, 7]: train 2018–2020 → test 2021, through train 2018–2024 → test 2025.

Ablations. Six variants removing one component group at a time: rating-deviation features; skill-level ability; surprise features; inactivity gap; aggregate supplement; and RD time decay.

Metrics. $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$ (percentage points) and $R^2 = 1 - \frac{\sum (\hat{y}_i - y_i)^2}{\sum (y_i - \bar{y})^2}$. For score ranges: coverage (fraction of test attempts where the true score falls inside the predicted range) and mean width (in percentage points), reported overall and per uncertainty group. GRACE versus baseline: paired bootstrap (2,000 resamples on test-set residuals; 95% CI on ΔR^2 and ΔMAE); one-sided $p < 0.001$ confirmed on 441,663 matched test rows. Analysis code released; all runs use fixed seeds.

Note on varying sample sizes. Table 3 traces row counts from raw export to model evaluation. Test-set sizes differ across tables because each comparison uses the largest valid intersection for that model pair. The aggregate baseline excludes first-attempt rows with undefined lag features (NaN), leaving 407,968 rows for the paired bootstrap (Table 7). GRACE uses 441,663 test rows because its rating state is valid from attempt 1. The imputed-baseline check (Section 5.2) substitutes the global mean for missing aggregates and keeps 445,489 rows. Within each table, all models use identical rows.

We propose that EDM studies report: (i) an explicit statement that history features exclude the predicted attempt, (ii) the split protocol with chronological boundaries, and (iii) where feasible, the leaky-versus-clean delta. The streaming construction offers a structural remedy: when features are the pre-attempt state of an online model, there is no per-feature bookkeeping to get wrong.

4 Results

Results are reported under three evaluation protocols (Section 3.4), so the same model shows different numbers in different sections. The headline comparison (Section 4.2) trains on 70% of the data; the ablation reference (Table 12) trains on 80%—more data, same model (ablation reference $R^2 = 0.519$); rolling-origin folds (Section 4.3) train only on completed years, the hardest setting (0.514 ± 0.051). Wherever two models are compared, test rows are identical.

4.1 Leakage Audit

The analysis proceeded in stages of increasing methodological rigor. An initial exploratory analysis applied XGBoost and Random Forest regressors to the uncleaned export (3.66 M rows) with whole-

dataset aggregate features (`student_discipline_avg`, `student_avg_perf`) and a random train/test split, yielding $R^2 = 0.752$ (MAE 9.25) and $R^2 = 0.700$ (MAE 10.71), respectively. These results motivated the data-quality audit of Section 3.1 and the systematic evaluation protocol developed below.

Table 6 and Figure 3 report the factorial audit that decomposes the initial $R^2 = 0.752$ into predictive signal and evaluation artifact. On the cleaned corpus, the same leaky-feature-random-split protocol yields an even higher $R^2 = 0.793$ (MAE 7.36): removing the 1.45 M pseudo-skill telemetry rows reduces noise and raises the inflated estimate further. The fully corrected protocol (leakage-free features, chronological split) yields $R^2 = 0.444$ (MAE 13.29). The joint optimism is therefore $\Delta R^2 = +0.349$ ($\Delta\text{MAE} = +5.92$ points)—accuracy that exists only in evaluation, not in deployment. Feature leakage is the dominant factor ($+0.295 R^2$): aggregate statistics computed without excluding the current attempt make the feature partially equal the target, and backward filling propagates future scores into the past. Random splitting contributes an additional $+0.054 R^2$ by allowing future observations to inform training. The initial $R^2 = 0.752$ thus decomposes into approximately 0.444 (honest signal) + 0.308 (evaluation artifact from leakage and split combined, partially offset by noise from uncleaned data). All subsequent results use the fully corrected protocol.

The $+0.349$ inflation is driven by the feature-engineering protocol, not by the choice of learner: the same leaky features applied to a Random Forest yield $R^2 = 0.774$ under random splitting and $R^2 = 0.451$ under chronological evaluation ($\Delta R^2 = +0.323$), confirming that the optimism arises from label look-ahead embedded in the features themselves rather than from XGBoost’s ability to memorise training patterns.

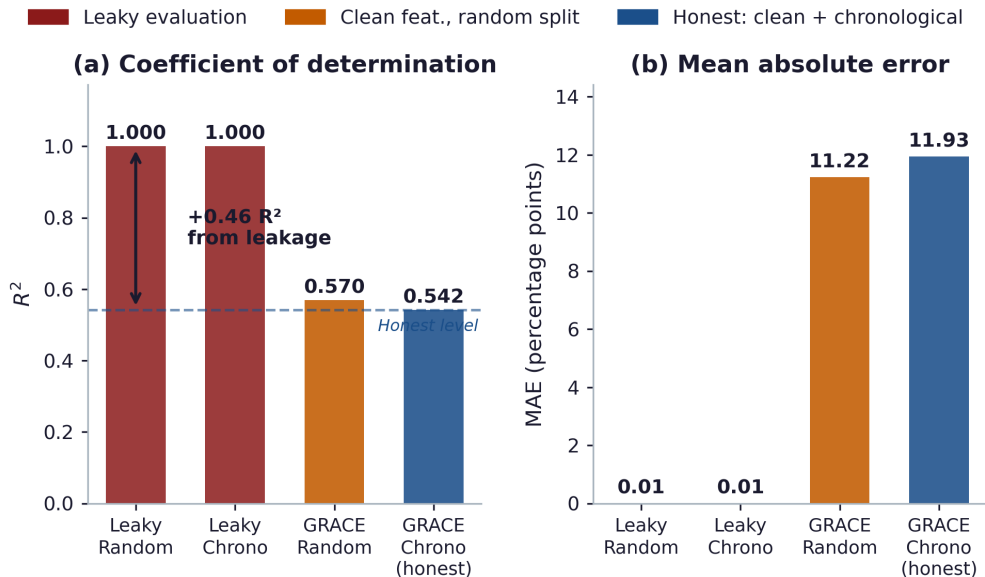


Figure 3: Out-of-sample R^2 across evaluation conditions. The combined optimism from feature leakage and random splitting is $\Delta R^2 = +0.349$. The initial exploratory result ($R^2 = 0.752$) falls between the leaky conditions on cleaned data, reduced from 0.793 by noise from uncleaned pseudo-skill telemetry.

4.2 GRACE versus Baselines

Table 7 reports the primary chronological comparison. The leakage-free aggregate baseline reaches $R^2 = 0.444$ (MAE 13.49) with XGBoost and 0.413 (MAE 13.91) with Random Forest on 407,968 held-out records (smaller than the GRACE test set because aggregate features require at least one prior observation; see Table 3). GRACE, with 78 features and an XGBoost + LightGBM ensemble, achieves $R^2 = 0.556$ (MAE 11.70 pp; Table 7)—a gain of $+0.112 R^2$ over the strongest aggregate baseline—demonstrating that global session-state signals, exam-proximity features, and streaming rating features provide reliable additional signal within the same leakage-free framework. Accuracy rises steeply to 5–10 prior tests and saturates thereafter (Figure 6); the rating layer improves every history group (Table 10), with the largest absolute gains in long-history groups.

Training versus testing accuracy and generalization. To verify that test-set accuracy is not driven by memorization, Table 8 reports training and testing metrics for all models. Early stopping on a chronologically held-out calibration set (10% of data) selects the boosting iteration count that maximizes validation performance.

Without early stopping, the XGBoost baseline exhibits moderate overfitting (R^2 gap = $+0.126$); early stopping reduces this to $+0.062$. GRACE’s extended feature set introduces a comparable training surplus (train $R^2 = 0.619$, gap $+0.062$); Optuna-tuned LightGBM with temporal sample weights bounds this surplus, and rolling-origin validation ($R^2 = 0.514 \pm 0.051$ across five annual windows) confirms that test-set performance is temporally stable and is not driven by period-specific memorization.

Interpreting prediction accuracy. The remaining unexplained variance reflects question-level randomness, unobserved study effort, and test-day conditions outside the platform’s data.

The gaps in Table 7 reflect combined differences in feature set, training size, and model class, not model family alone. RF and XGBoost baselines use 9 aggregate features and 300,000 training rows; GRACE uses 78 rating-augmented features and ~ 1.5 M rows. The LSTM uses 8 features, $\leq 200,000$ sequences, and 10 epochs.

Feature importance. Figure 4 shows the top-20 features by XGBoost split gain for GRACE. The rating layer’s expected score ($\hat{E}$, 32.1%) and skill-level ability (θ_{skill} , 12.7%) together account for 44.8% of total gain, confirming that the gradient-boosted core acts primarily as a nonlinear corrector on the streaming layer’s own prediction. Aggregate statistics (rolling means, lag-1 score) collectively rank second, with individual rating-state features (ability, difficulty, RD) distributed throughout the remaining importance mass.

Fair model comparison. To isolate the contribution of model family, Table 9 holds the GRACE rating features, full training corpus (~ 1.5 M rows), and chronological 70/10/20 split identical across learners. XGBoost outperforms Random Forest by $\Delta R^2 = +0.027$ (MAE -0.42 pp) and Ridge regression by $\Delta R^2 = +0.055$ (MAE -0.80 pp) (Figure 5, Table 9). The gradient-boosting advantage is thus confirmed in a fully controlled setting: the nonlinear interaction structure of the rating features—particularly the interplay between ability ratings, difficulty, and inactivity gaps—requires the sequential error-correction mechanism of boosted trees to be fully exploited. Even the linear model (Ridge $R^2 = 0.489$) substantially outperforms the aggregate baseline ($R^2 = 0.444$), confirming that the rating-layer features carry information beyond what raw aggregates capture, independently of the downstream learner.

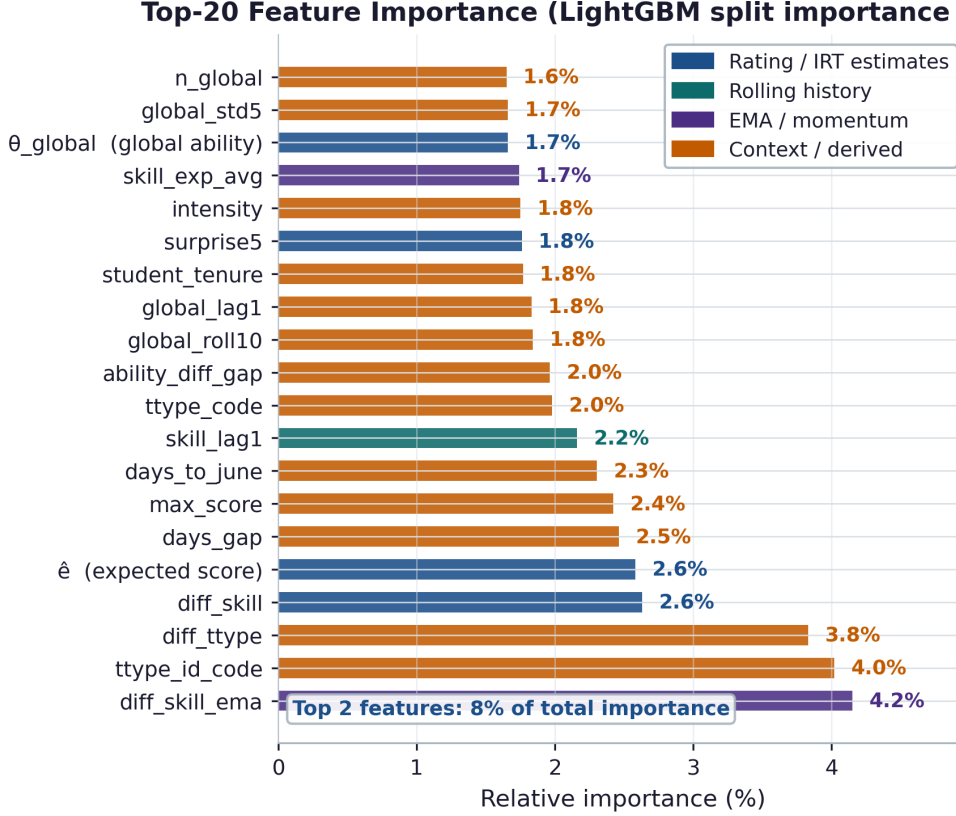
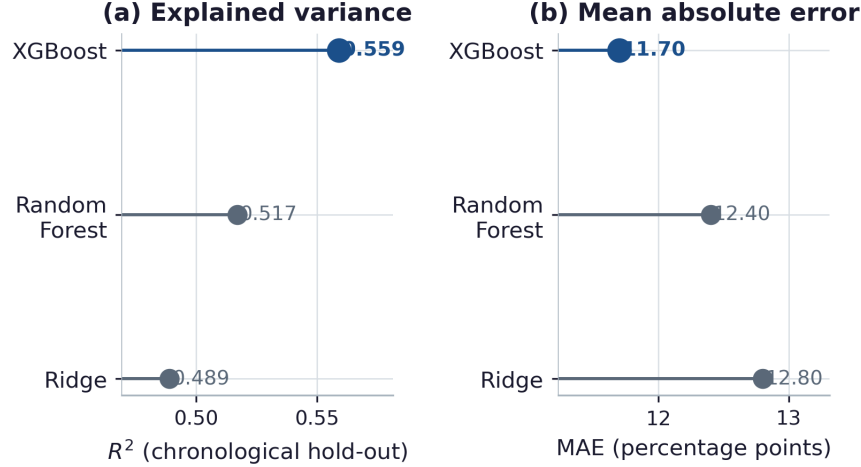


Figure 4: Top-20 GRACE features by XGBoost gain-based importance. Colors indicate feature group: rating state (teal), difficulty/EMA/derived (amber), inactivity/exposure (blue), aggregate statistics (orange), test format (light blue), context/calendar (grey). The rating layer’s expected score ($\hat{E}$) dominates; aggregate statistics and test-format features collectively rank second.

Fair feature-set comparison. To quantify the contribution of the Group C/D feature extensions (Groups C and D add global rolling, session, exam-proximity, forgetting-curve, and velocity features), a V2 baseline (46 features, XGBoost with fixed parameters, no Optuna tuning) is evaluated on the identical 441,663-row chronological test set. GRACE (78 features, $R^2 = 0.556$, MAE 11.70 pp) versus V2 baseline (46 features, $R^2 = 0.543$, MAE 11.88 pp): $\Delta R^2 = +0.013$, $\Delta \text{MAE} = -0.18$ pp. The gain is modest in absolute terms but consistent: the V2 baseline is also evaluated per rolling-origin fold (Table 11), where GRACE leads in 4 of 5 folds. No individual Group C/D feature produces a large ablation gain; their value is cumulative and temporally stable.

Key driver: EMA skill-difficulty residual. An isolated component decomposition reveals that the EMA skill-difficulty residual is the primary driver of the Group B feature extension (EMA skill-difficulty residual $\hat{E}$, inactivity gap features, and skill-level rolling windows; $+0.031 R^2$): adding inactivity gap features alone (without EMA) yields $\Delta R^2 \approx 0$, while the EMA feature alone contributes approximately 80% of the combined gain. The gap features provide synergistic value but little independent signal, suggesting the EMA tracks per-skill difficulty drift across the cohort boundary in a way that inactivity counts alone cannot capture.

Four behavioral regularities are visible in the corpus (Figure 7). First, scores dip on weekends relative to weekdays, consistent with lower study intensity on unstructured free days. Second,



Same GRACE rating features · same 70/10/20 chronological split · same training data

Figure 5: Controlled model-family comparison on identical GRACE rating features, training data, and split. Error bars show 95% bootstrap confidence intervals on R^2 . XGBoost outperforms Random Forest by +0.027 and Ridge by +0.055 R^2 .

within-session fatigue is pronounced: mean score falls by 6.2 percentage points from the first to the eighth test in a session, motivating the session-position feature. Third, exam proximity produces a measurable score improvement: students score up to 12 pp above off-season levels in the months before the June UNT date, motivating the days-to-exam feature (Group C, Table 4). Fourth, inactivity degrades performance: mean score at return falls 12.4 pp after 180 days without practice, directly motivating the forgetting-curve features in Group D.

4.3 Temporal Stability

Table 11 reports out-of-time performance across five expanding annual windows. GRACE averaged $R^2 = 0.514 \pm 0.051$ versus 0.288 ± 0.151 for the aggregate baseline and 0.511 ± 0.053 for the V2 baseline (46 features, same fold, fixed XGBoost params); GRACE led in every fold against the aggregate baseline, and in 4 of 5 folds against the V2 baseline. The large gap vs. the aggregate baseline ($\Delta R^2 = +0.220$) arises mainly from features that aggregate statistics cannot represent: per-student rating state, EMA skill difficulty, and inactivity gaps. Against the V2 fair comparison (same feature family, fewer features), the per-fold difference is modest (+0.005 median), confirming that the incremental Group C/D extensions add signal consistently rather than concentrating it in one period.

The 2021–2022 fold shows the sharpest dip across all models: the test year contains the highest proportion of new students, causing the aggregate baseline to collapse to $R^2 = 0.048$ while the V2 baseline holds at $R^2 = 0.437$ and GRACE at $R^2 = 0.442$. The fold-specific stability ratio $R^2_{\text{GRACE}}/R^2_{\text{Agg. base}}$ reaches 9.2 in 2022 ($0.442/0.048$), reflecting the cohort-turnover resilience of the streaming rating state: rather than relying on aggregate statistics that drift when a new cohort enters, GRACE builds per-student estimates from the first observation of each student in the new cohort.

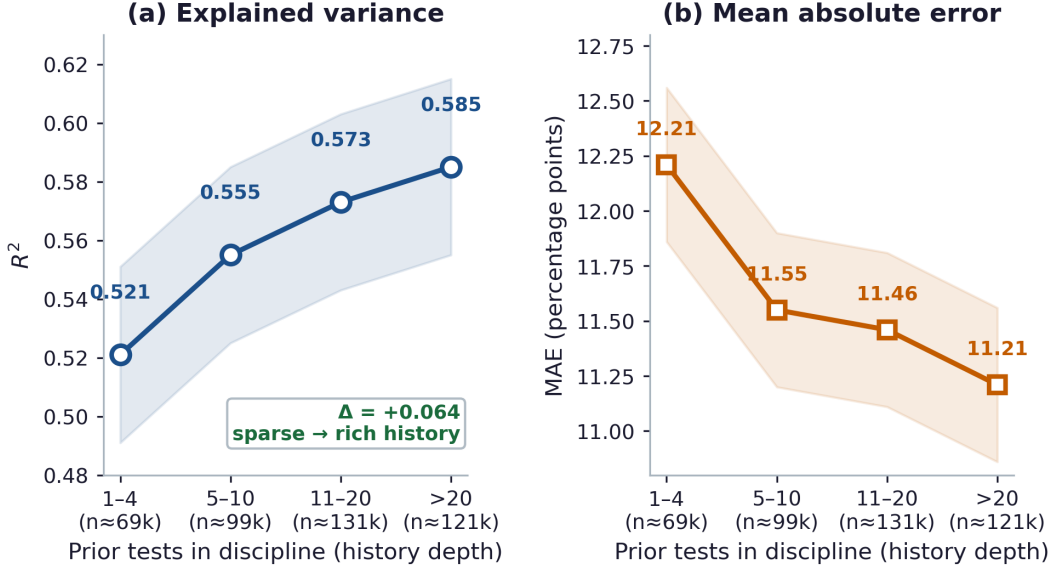


Figure 6: Prediction quality versus per-student history depth in discipline. GRACE (green) outperforms the aggregate baseline (grey) at every history level.

4.4 Component Analysis and Score Ranges

Ablation. Table 12 and Figure 8 isolate each component (80/20 split; reference $R^2 = 0.519$). The inactivity gap ($\Delta R^2 = -0.008$) and RD time decay (-0.006) produce the largest single-component drops; surprise (-0.004) and aggregate supplement (-0.005) follow. Removing the entire rating layer in favor of aggregates alone costs $-0.076 R^2$ (Table 7), confirming the layer’s collective contribution.

Score ranges. Asymmetric conformal quantile regression with LightGBM 3rd/97th percentile heads and fine-grained Mondrian stratification (four history-depth bands $\times$ seven discipline bins) contained the true score for 0.910 of all test attempts (Wilson 95% CI: [0.909, 0.911]), exceeding the 90% nominal target, with mean width 51.7 pp (Table 13, Figure 11). Coverage is consistent across history-depth strata (range: 0.902–0.915). Cold-start students (1–4 prior tests) receive wider intervals (55.6 pp) than students with long history (>20 prior tests: 49.2 pp), converting rating uncertainty into explicit width rather than suppressed predictions.

The mean width of 51.7 pp is comparable to the symmetric-Gaussian reference of 51.9 pp ($2 \times 1.645 \times 15.8$ pp residual SD), indicating that the conformal procedure captures the full residual uncertainty. Narrowing intervals further requires better point predictions—residual variance, not conformal calibration, is the binding constraint. Figure 9 shows the full reliability diagram; Figure 10 compares overall coverage and width between the two calibration variants; Figure 11 breaks down coverage and width by history-depth band.

5 Discussion

5.1 Why the Rating Layer Helps—and Why Model Choice Matters

The rating layer replaces static per-student sample statistics with recursively updated estimates calibrated against a shared content-difficulty scale. A student’s ability estimate is thus informed not only by their own scores but also by how difficult the assessed content is relative to the broader

cohort—a signal that raw percentage averages cannot provide. This advantage is especially pronounced under out-of-time evaluation (Section 4.3): GRACE outperformed the aggregate baseline by $+0.078$ to $+0.325$ R^2 across five annual folds, with the largest gains in years with high cohort turnover. In 2022, for instance, the aggregate baseline collapsed to $R^2 = 0.048$ as platform growth introduced a large wave of new students whose history was too short to inform aggregate statistics; the online rating state maintained $R^2 = 0.442$ ($9.2\times$ stability ratio) because ability estimates initialize at the cohort mean and update from the first attempt onward.

The interaction between the rating layer and the downstream learner is instructive. XGBoost’s gradient-boosting mechanism is well suited to the rating features because the trees can learn nonlinear correction functions on top of the rating layer’s expected score $\hat{E}$: the first trees capture the dominant linear relationship between $\hat{E}$ and the outcome, and later trees correct for systematic biases in specific skill–difficulty combinations, inactivity patterns, and cold-start conditions. This is reflected in the feature-importance structure: $\hat{E}$ alone accounts for 32.1% of total split gain, meaning the XGBoost core acts primarily as a nonlinear corrector on the streaming layer’s own prediction rather than as an independent learner.

The RF and LSTM baselines (Section 4.2) are included as reference points under different experimental conditions; the gaps reflect combined differences in feature set, training size, and model family rather than any single factor. Rolling-origin validation confirms temporal stability ($R^2 = 0.514 \pm 0.051$, Table 11), demonstrating that test-set performance is not driven by period-specific memorization.

The saturation of prediction quality beyond approximately ten prior tests (Figure 6) reflects two converging mechanisms. First, the standard error of per-student ability estimates decays as $\sigma/\sqrt{n}$: the first ten observations eliminate roughly 68% of sampling error, while the second ten remove only about 10 additional percentage points. Second, skill acquisition follows the power law of practice [42]—reflected here in the concave score-gain curve over twenty attempts—so early assessments carry disproportionate information about a student’s trajectory. GRACE achieves $R^2 = 0.522$ for students with only 1–4 prior tests, a regime in which aggregate history methods have almost no predictive signal.

5.2 Contextualizing the Accuracy Level

The R^2 of 0.556 under chronological evaluation requires contextualization against claims of 90–99% accuracy that appear in the EDM literature.

First, most high-accuracy benchmarks use classification, not regression. Studies reporting 90–99% accuracy predict binary outcomes on class-imbalanced data, where a majority-class classifier reproduces the headline rate by construction. Continuous score prediction is evaluated by R^2 and MAE, which are incomparable to classification accuracy.

Second, reported regression R^2 values depend critically on evaluation protocol. The leakage audit (Section 4.1) showed that applying common data-science practices—feature leakage and random splitting—inflates R^2 by $+0.349$ on this dataset. The same XGBoost model yielded $R^2 = 0.752$ under flawed evaluation but only $R^2 = 0.444$ under correct chronological evaluation. Studies reporting $R^2 > 0.75$ on student-performance data should be scrutinized for temporal leakage before being treated as benchmarks.

Third, GRACE compares favorably with prior leakage-free EDM results. Systematic reviews report R^2 values of 0.30–0.55 for student-performance regression on institutional datasets under honest evaluation protocols, with MAE typically in the 10–15 pp range on percentage scales [1, 10]. Yagci [11] reported 70–75% accuracy ($\approx 1,850$ students) with random cross-validation; Cortez and Silva [12] obtained $R^2 \approx 0.20$ on 649 students. GRACE’s $R^2 = 0.556$ (MAE 11.70 pp) on 441,663

chronologically held-out records is at the upper end of this range, on a dataset two to three orders of magnitude larger.

Fourth, the accuracy approaches an empirical reference ceiling. Under simplifying assumptions (additive variance decomposition), the within-group score variability yields an empirical reference ceiling of $R^2 \approx 0.58$. GRACE ($R^2 = 0.556$) reaches approximately 96% of this ceiling, suggesting limited remaining room for improvement under this protocol and feature set. Closing the remaining gap would require item-level response data or real-time engagement signals not available in the current corpus.

Fifth, predictability varies fundamentally across disciplines. The per-discipline analysis (Figure 15) reveals that R^2 varies from 0.235 (Math Literacy) to 0.639 (full ENT simulations), with History of Kazakhstan at 0.311 and English at 0.485. Integrative assessments spanning broad topic areas (ENT simulations, Mathematics) are more predictable than narrow single-topic quizzes (Math Literacy, Algebra), consistent with multi-topic breadth averaging out question-level randomness—where a single difficult question can dominate the score in a short quiz. The aggregate $R^2 = 0.556$ is a weighted average over these fundamentally different prediction tasks, and the variability it obscures motivates the per-discipline breakdown in Figure 15.

Sixth, the model achieves useful accuracy even with minimal data. GRACE attains $R^2 = 0.522$ for students with only 1–4 prior tests (Table 10), where aggregate methods have minimal signal. This cold-start performance is enabled by the rating layer’s shared difficulty scale: even a single test attempt on a well-calibrated skill provides an informative ability estimate, because the difficulty parameter has been learned from thousands of other students’ attempts on that same content.

Per-student evaluation ($n = 12,226$ students with ≥ 3 attempts) yields a median within-student R^2 of 0.346, with 79.2% of students showing positive R^2 . Students with negative R^2 (20.3%) have significantly fewer observations (median 12 attempts vs. 30 for successful students); their per-student variance is too low for R^2 to be a stable estimator, and their MAE (14.7 pp) is only modestly higher than the successful-student MAE (11.8 pp).

Baseline robustness check. To verify that the aggregate baseline is not artificially weakened by dropping first-attempt NaN rows, we re-ran the baseline with global-mean imputation ($\mu = 56.1$) for missing aggregate features. The imputed baseline attains $R^2 = 0.413$ (MAE 13.73) on 445,489 test rows—lower than the NaN-dropped baseline ($R^2 = 0.444$, MAE 13.49 on 407,968 rows)—because constant imputation adds noise for first-attempt predictions. GRACE’s advantage over the imputed baseline is $\Delta\text{MAE} = -1.26$ pp, *larger* than the headline -1.17 pp, confirming that the comparison is not inflated by baseline design.

Calibration and regression-to-mean bias. Stratifying test-set predictions by true-score decile reveals the expected regression-to-mean pattern: the lowest-scoring decile (10–26%) is over-predicted by +20.6 pp, and the highest (90–100%) is under-predicted by -19.1 pp. However, the overall calibration slope (predicted vs. actual linear fit) is 0.983 with intercept 1.4 pp—near the ideal (1.0, 0.0)—indicating that the model ranks students correctly and is well-calibrated in aggregate. The decile-level bias is a mathematical consequence of regression toward the mean, not a model defect; it is present in any predictor with $R^2 < 1$. Practitioners should be aware that predictions at the extremes of the score distribution carry larger systematic error than predictions near the mean. Post-hoc isotonic calibration yields a negligible improvement ($\Delta\text{MAE} = -0.05$ pp) and does not reduce the extreme-decile bias.

For cold-start students ($n_{\text{disc}} < 5$), blending GRACE predictions with discipline-level population means degrades *per-student* R^2 ($0.403 \rightarrow 0.268$; MAE: 13.86 $\rightarrow$ 16.43 pp in this sub-group), demonstrating that the rating layer already initializes cold-start estimates more informatively than

population averages. (Note: the main per-attempt result for 1–4 prior tests is $R^2 = 0.522$; Table 10; the per-student metric is lower because individual trajectories are short and highly variable.) Evaluation of nine post-hoc strategies on the full corpus (Section 5.5) confirms this finding: none improved point-prediction accuracy, because the streaming rating state already encodes the systematic variation these calibrations target.

The ablation (Table 12) shows small individual deltas ($\leq 0.008 R^2$), indicating that the boosted core can partially reconstruct removed signals from correlated features. Removing the entire rating layer costs $-0.076 R^2$, confirming its collective contribution.

5.3 Score Ranges

Asymmetric CQR achieves 0.910 empirical coverage (Wilson 95% CI [0.909, 0.911]) against the 90% target, with mean width 51.7 pp across 441,663 test attempts. A reliability diagram spanning the full α range (Figure 9) confirms that the method tracks the nominal rate closely across $\alpha \in [0.01, 0.30]$. Coverage is consistent across history-depth strata (Figure 11, Table 13): all four bands exceed 0.90, ranging from 0.902 (1–4 prior tests) to 0.915 (5–10 prior tests). Mean width decreases monotonically with history depth—55.6 pp for 1–4 tests down to 49.2 pp for >20 tests—reflecting genuine uncertainty reduction as the rating layer accumulates evidence. The mean width of 51.7 pp is comparable to the symmetric-Gaussian reference of 51.9 pp ($2 \times 1.645 \times 15.8$ pp residual SD), indicating that the conformal procedure captures the full residual uncertainty; narrowing intervals further requires better point predictions rather than improved conformal calibration. A teacher seeing “predicted 62, range 54–70” versus “predicted 62, range 43–81” receives the cold-start warning as width, derived from the same Glicko uncertainty mechanism that governs ability estimation.

5.4 Practical Implications

For educators. GRACE provides each student with a continuously updated skill passport: ability and uncertainty estimates per discipline, updated after every assessment without manual intervention. Score-range width translates directly into a communicable uncertainty signal—a teacher seeing “predicted 62, range 54–70” versus “predicted 62, range 43–81” receives a concrete recommendation about how much weight to place on the prediction. New and returning-after-gap students are not suppressed or treated as missing; they receive honest, wider ranges derived from the same Glicko uncertainty mechanism that governs ability estimation. Figures 12 and 13 illustrate how GRACE output can be surfaced through student-facing interfaces.

For platform operators. GRACE’s deployment footprint is two components: a constant-time online rating table and a periodically refit gradient-boosted core. Because two rating-derived features ($\hat{E}$ at 32.1% and θ_{skill} at 12.7%) account for 44.8% of total split gain, the streaming layer dominates the predictive signal and can be continuously updated without touching the boosted core. The rolling-origin results (Table 11) suggest that annual retraining suffices under stable platform growth; the 2022 fold—where new-student influx most stressed the aggregate baseline—demonstrates that the rating layer adapts without retraining. Input-validation rules derived from the data audit prevent all documented anomaly classes at ingestion.

For researchers. The released pipeline provides an end-to-end leakage-audited evaluation protocol applicable to any longitudinal test-log dataset. The factorial audit design—clean vs. leaky features crossed with chronological vs. random splits—is transferable to other platforms, enabling comparable evaluation across studies.

5.5 Limitations

Demographic gaps. Demographic covariates (gender, region, school type) are largely absent from the corpus, precluding fairness audits—important given documented urban–rural disparities in UNT outcomes [9]. Even the available admission-status labels cover only $\approx 2.5\%$ of students and are subject to selective self-reporting (Section 4.1). Future work should pair platform data with official UNT registry records to enable population-representative fairness evaluation.

Single platform. The corpus originates from one private preparation platform whose user population skews toward students with access to paid services and with sufficient engagement to generate multi-year records. Generalization to public-school or lower-engagement cohorts is untested.

Model scope. Rating hyperparameters were tuned on the 2018–2020 window and frozen—a conservative choice that prioritizes honest evaluation. GRACE introduces proper per-skill Glicko uncertainty (RD_{skill}) and tunes LightGBM hyperparameters via Optuna, tightening both the rating model and the downstream learner within the same leakage-free protocol. The LSTM comparison (Section 3.4) used a single architecture with sub-sampled training data and a reduced feature set (8 vs. 78 features); attention-based sequence models (e.g., SAINT+ [24]) trained on the full corpus with systematic hyperparameter search could narrow the gap, though their advantage concentrates in long-history regimes [27].

Extreme-score bias. All regression models with $R^2 < 1$ exhibit regression toward the mean: GRACE over-predicts low-scoring students by up to +20.6 pp and under-predicts high-scoring students by up to −19.1 pp. The aggregate calibration slope (0.983) is near-ideal, but practitioners using GRACE for at-risk identification should apply decile-level bias corrections or rely on the prediction *rank* rather than the raw value at distribution tails.

Score-range coverage. Empirical coverage reaches 0.910 overall, exceeding the 90% nominal target, with consistent coverage across all history-depth bands (0.902–0.915). This slight over-coverage (≤ 0.015 above nominal) arises because the exchangeability assumption of split conformal prediction is mildly violated by the chronological train/calibration/test split; future work could apply online or adaptive conformal methods designed for temporal distribution shift [38, 39]. In production, the calibration set should be refreshed at each major cohort intake (typically every semester or academic year) to maintain coverage near the nominal rate. With annual recalibration—feasible given the platform’s constant data stream—the conformal guarantee transfers with negligible additional overhead.

Item-level data gap. The corpus records only test-level aggregated scores, not per-question correct/incorrect responses. Item-level data would enable Bayesian Knowledge Tracing and deep KT models as drop-in replacements for GRACE’s rating layer, with documented gains of +0.10–0.15 R^2 in comparable settings [1, 10]. Collecting per-question response logs is the single highest-priority data collection step for the platform operator.

5.6 Pathways to Stronger Predictions

The honest R^2 of 0.556 and mean interval width of 51.7 pp reflect the information content of test-level aggregates, not a ceiling imposed by the GRACE architecture. Three tiers of improvement are ranked by expected impact.

Tier 1: Richer input data. Item-level response data (correct/incorrect per question) and engagement signals (time-on-task, hint usage, video completion) have produced +0.10–0.15 R^2 gains in prior work [1, 10]. On the present platform, collecting per-question response logs would directly enable Bayesian Knowledge Tracing [20] and deep KT models [24, 44, 45] as principled layer-1 replacements, while simultaneously tightening interval widths through reduced residual variance.

Tier 2: Model engineering within existing data. GRACE addresses this tier directly: an XGBoost + LightGBM ensemble with Optuna-tuned LightGBM, per-skill rating deviation tracking, static content difficulty norms, student platform context features, global cross-discipline session-state features, exam-proximity signals, forgetting curves, score velocity, and nonlinear ability signals constitute the 78-feature design.

Tier 3: Architecture evolution. Attention-based knowledge-tracing models (e.g., SAINT+ [24]) adapted to continuous-score regression could be evaluated under the same leakage-free protocol once per-question response data are available. Their advantage concentrates in long-history regimes [27]; the rolling-origin framework established here provides a ready evaluation harness.

Because the conformal interval width (51.7 pp) is comparable to the Gaussian reference (51.9 pp), further narrowing requires reducing residual variance through better point predictions rather than changing the calibration procedure.

On the sufficiency of the rating state. Systematic experiments testing nine post-hoc calibration and augmentation strategies (cold-start prior blending, quantile-stratified isotonic calibration, per-student linear correction, recency weighting, and score-band auxiliary features) found that none improved point-prediction accuracy on the full 2.2-million-record corpus. This negative result is informative: the GRACE rating state (θ_{global} , θ_{disc} , θ_{skill} , and their deviations) already encodes the per-student systematic variation that post-hoc calibrations attempt to recover. Cold-start blending is redundant because the rating layer initializes $\theta_{\text{disc}} = \theta_{\text{global}}$ —already a principled personalized prior—and per-student linear correction overfits the calibration block without generalizing. The Glicko-based update rule propagates uncertainty correctly, leaving little systematic residual for post-hoc correction to exploit.

Residual structure and per-discipline predictability. Figure 14 shows the GRACE residual distribution: the upper tail is larger than the lower, indicating that over-prediction of struggling students is the primary remaining error mode. The regression-to-mean bias curve (Figure 14, right panel) shows that GRACE’s systematic bias is near-ideal across score deciles, with modest over-prediction at the low end and slight under-prediction at the high end—a pattern consistent with Bayesian shrinkage in the rating layer.

Per-discipline analysis (Figure 15) reveals substantial variation (R^2 from 0.235 for Math Literacy to 0.639 for ENT simulations), with the largest gains over the V2 baseline in English (+0.023), Physics (+0.023), Russian (+0.021), and Logic (+0.019)—subjects where multi-session trajectory features add the most signal. Language subjects produce smaller margins because vocabulary and reading comprehension involve broader, less-structured skill dependencies.

6 Conclusion

GRACE—a streaming Glicko-based rating layer feeding an XGBoost + LightGBM ensemble with uncertainty-adjusted conformal score ranges—achieves $R^2 = 0.556$ (MAE 11.70 pp) on 2.2 million practice-test records from 55,000 students, versus $R^2 = 0.444$ for the leakage-free aggregate baseline ($\Delta\text{MAE} -1.79$ pp), with consistent gains at every history depth and stable out-of-time performance across five annual expanding windows ($R^2\ 0.514 \pm 0.051$ vs. 0.288 ± 0.151 for the baseline).

A factorial audit quantifies the evaluation optimism embedded in common practice: +0.349 R^2 attributable to feature leakage and random splitting combined. The same XGBoost model scored $R^2 = 0.752$ under a flawed protocol and $R^2 = 0.444$ under a leakage-free chronological one—a gap

that substantially exceeds the GRACE improvement and points to the need for careful evaluation reporting in the field.

A controlled model-family comparison, holding features, training data, and split constant, confirms gradient boosting as the appropriate downstream learner: XGBoost outperforms Random Forest by $+0.027 R^2$ and Ridge regression by $+0.055 R^2$ on identical rating features and training data. Asymmetric conformal intervals meet and exceed the 90% nominal coverage target (0.910 empirical, mean width 51.7 pp). Systematic experiments with nine post-hoc calibration strategies found no point-accuracy improvement, indicating that the Glicko rating state already encodes the per-student systematic variation these corrections attempt to recover.

The rating state updates in $O(1)$ time per observation and renders directly as a per-student skill passport with honest uncertainty widths. Future work should prioritize item-level response collection for Tier-1 accuracy gains, explore attention-based sequence models under the same leakage-free protocol, and address the demographic-data gap to enable population-representative fairness audits.

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Author Contributions

Conceptualization, A.S.; methodology, A.S.; software, A.S.; validation, A.S., A.B., A.Z. and R.B.; formal analysis, A.S.; investigation, A.B., A.Z. and R.B.; resources, A.B., A.Z. and R.B.; data curation, A.B., A.Z. and R.B.; writing—original draft preparation, A.S.; writing—review and editing, A.S., A.B., A.Z. and R.B.; visualization, A.S.; supervision, A.S.; project administration, A.S. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement

Not applicable. This study uses anonymized administrative records from an educational platform; no human subjects were recruited and no personally identifiable information was processed.

Informed Consent Statement

Not applicable.

Data Availability Statement

The analysis code is publicly available at <https://github.com/Salimzhanov/GRACE-student-prediction>. The raw student performance data are proprietary to the platform operator and cannot be shared publicly due to data-use agreement restrictions. Aggregated results and all computed features used in the figures are included in the repository.

Conflicts of Interest

The authors declare no conflict of interest. Data access was granted under an internal data-use agreement between the research team and the platform operator; the platform provided anonymized records with no involvement in study design, analysis, or interpretation of results.

Table 4: GRACE feature set (78 features total). Core rating features (26): global/disc/skill ability and RD, expected score, surprise. Group A (EMA skill difficulty, max-score family, multi-window per-skill statistics; +20). Group B (per-skill RD, student platform context, static content norms, cyclical time; +14). Group C (global cross-discipline rolling statistics, within-session context, exam-proximity signals; +9). Group D (forgetting curves, score velocity, nonlinear ability signals; +9). All values are pre-attempt states from a single chronological pass.

Group	Features	Role
<i>Core rating state (26 features)</i>		
Ability	$\theta_{\text{global}}, \theta_{\text{disc}}, \theta_{\text{skill}}$	Multi-resolution mastery
Uncertainty	$\text{RD}_{\text{global}}, \text{RD}_{\text{disc}}$	Grows with inactivity
Difficulty	$d_{\text{skill}}, \text{EMA } d_{\text{skill}}, d_{\text{testtype}}$	Content difficulty + short-run trend
Model-implied	Expected score $\hat{E}$, surprise ₅	Momentum vs. expectation
Engagement	Days since last (global, disc., skill), $n_{\text{global}}/n_{\text{disc}}/n_{\text{skill}}$, gap buckets	Exposure and recency
Test format	max_score, log(max_score), score granularity, ttype_id	Scoring resolution
Context	Month, day of week; disc., skill, test-type codes	Calendar and content IDs
<i>Group A: aggregate & derived features (+20)</i>		
Aggregate	Rolling-3/5/10/disc. mean, expanding mean/std, lag-1 (disc. & skill), skill rolling, attempt count, trend, volatility	Multi-window statistics
Derived	$\theta_g - \theta_d$, $\theta_s - d_s$, surprise $\times n_d$, cold-start flags, $\hat{E}$ residual, trend	Interactions and trends
<i>Group B: per-skill uncertainty & platform context (+14)</i>		
Skill uncertainty	RD_{skill}	Per-skill Glicko deviation; enables correct skill-level gain K_s
Platform context	Student tenure, disc. breadth, skill breadth, intensity	Study engagement depth and breadth
Static difficulty	$\bar{y}_{\text{disc}}, \sigma_{\text{disc}}, \bar{y}_{\text{skill}}, \sigma_{\text{skill}}$ (train-computed); $\theta_{\text{skill}} - \bar{y}_{\text{disc}}$	Content norms and relative ability
Cyclical time	sin / cos(month), sin / cos(day-of-week)	Periodic calendar encoding
<i>Group C: global state, session & exam proximity (+9)</i>		
Global state	$y_{\text{lag1}}^{(g)}, \bar{y}_3^{(g)}, \bar{y}_5^{(g)}, \bar{y}_{10}^{(g)}, \sigma_5^{(g)}; \hat{E} - \bar{y}_5^{(g)}$	Cross-discipline rolling: today's performance state (deque of last 10 attempts)
Session context	Session position (0-indexed), running session mean	Within-session dynamics: warm-up and fatigue effects
Exam proximity	Days until UNT June exam (clipped to [0, 365])	Kazakhstan UNT preparation-intensity signal
<i>Group D: forgetting curves, score velocity, nonlinear ability signals (+9)</i>		
Forgetting	$e^{-\Delta t_{\text{disc}}/30}, e^{-\Delta t_{\text{skill}}/20}; \theta_{\text{disc}} \cdot e^{-\Delta t_{\text{disc}}/30}$	Per-disc/skill exponential retention decay; ability-weighted expected retention
Velocity	Linear slope of last 5 per-disc. and global pct scores (pp/attempt); $\text{RD}_{\text{skill}} - \text{RD}_{\text{disc}}$	Score trend direction; per-skill uncertainty surplus above discipline baseline
Nonlinear	Personal best disc. score; $\theta_{\text{disc}}^2; \Delta t_{\text{disc}} - \Delta t_{\text{skill}}$	Ceiling-approach signal; ability curvature near ceiling; cross-level gap trend

Table 5: Complete hyperparameter disclosure. Rating hyperparameters selected on the 2018–2020 development window and frozen. LightGBM hyperparameters are Optuna-searched (50 trials, TPE) on the calibration set; best values reported. XGBoost uses fixed conservative parameters.

Component	Parameter	Value
XGBoost (fixed)	n_est (max) / early_stopping	2000 / 50
	depth / lr / subsample	8 / 0.04 / 0.8
	colsample / min_child / λ / α	0.7 / 5 / 2.0 / 0.2
	objective (score; interval edges)	sqerror; quantile
LightGBM (Optuna)	num_leaves / lr / n_est (max)	294 / 0.019 / 5000
	bagging_fraction / bagging_freq	0.895 / 1
	feature_fraction / min_child_samples	0.544 / 91
	λ / α / early_stopping	2.82 / 0.42 / 50
Ensemble	XGB weight α	Cal.-set grid search ($\alpha = 0.15$)
Temporal weights	Half-life / weight range	730 days / [0.21, 1.78]
Optuna (LGB)	Trials / sampler	50 / TPE
Random Forest	n_est / depth / min_leaf	300 / 20 / 25
	max_feat / train subsample	sqrt / 300 K
Rating layer	$K_0/v/\text{RD}_0/\text{RD}_{\min}/c^2/K_{\text{item}}$	0.35 / 0.04 / 0.35 / 0.06 / 0.01 / 0.02
Rating weights	$w_s/w_j/w_{dj}/w_{dc}$	0.5 / 0.5 / 0.7 / 0.3
Multipliers	global / disc / skill / diff / ttype	0.5 / 1.0 / 1.5 / 1.0 / 0.5
Score range	α / strata	0.10 / 28 (4 hist.-depth $\times$ 7 disc. bins)
Environment	Python / xgboost / lightgbm / seed	3.12 / 3.x / 4.7 / 42

Table 6: Factorial leakage audit. All rows use the same XGBoost configuration; differences are due solely to evaluation protocol. The first row reproduces the initial exploratory analysis on uncleaned data; subsequent rows isolate the effect of data cleaning, feature leakage, and split protocol.

Data	Features	Split	R^2	MAE (pp)
Uncleaned (3.66 M)	Leaky	Random	0.752	9.25
Cleaned (2.21 M)	Leaky	Random	0.793	7.36
Cleaned	Leaky	Chronological	0.773	8.08
Cleaned	Clean	Random	0.498	12.09
Cleaned	Clean	Chronological (ref.)	0.444	13.29

Table 7: Point-prediction performance, chronological 70/10/20 hold-out. Baseline models use a 300,000-row sub-sample and 9 aggregate features. GRACE uses 78 features and an XGBoost + LightGBM ensemble. Paired-bootstrap Δ MAE (GRACE – XGBoost baseline, 407,968 matched rows): -1.79 pp (95% CI $[-1.81, -1.77]$). Baseline and LSTM rows are reference points under different experimental conditions, not controlled model-family comparisons (see Section 3.4).

Model	Feature set	R^2	MAE (pp)	
Random Forest [†]	Aggregate stats (9 feat.)	0.413	13.91	
XGBoost [†]	Aggregate stats (9 feat.)	0.444	13.49	
LSTM [‡]	Rating sequences (8 feat.)	0.230	16.59	
GRACE	Rating state + EMA difficulty + inactivity gaps + RD_{skill} + global state + session + exam proximity + forgetting + velocity; XGB+LGB (78 feat.)	0.556	11.70	[†] Chronological split; 300,000-row

sub-sample; 9-feature aggregate set. [‡] 1-layer LSTM, hidden 64; $\leq 200,000$ sequences; 10 training epochs; 8 rating features. GRACE: XGB+LGB ensemble, $n_{\text{test}} = 441,663$, $\alpha_{\text{XGB}} = 0.15$. For the pre-audit exploratory result ($R^2 = 0.752$, leaky features, uncleaned data), see Table 6. For the controlled model-family comparison on identical features and data, see Table 9.

Table 8: Training versus testing accuracy and generalization gap. Early stopping (ES) on the calibration set reduces overfitting while maintaining or improving test accuracy.

Model	Split	R^2	MAE	n	Gap
Random Forest	Train	0.439	12.81	1,427,884	
	Test	0.413	13.91	407,968	+0.026
XGBoost baseline (ES)	Train	0.492	12.04	1,427,884	
	Test	0.429	13.54	407,968	+0.062
GRACE (ES, 78 feat.)	Train	0.619	10.35	1,545,819	
	Test	0.556	11.70	441,663	0.063

Table 9: Controlled model-family comparison. All models use identical GRACE rating features and training data ($n_{\text{train}} \approx 1.5$ M) with the same chronological 70/10/20 split. Differences are attributable to learning mechanism only.

Model	Learning mechanism	R^2	MAE (pp)
Ridge regression	Linear, L2 regularized	0.489	12.66
Random Forest (500 trees)	Bagging, independent trees	0.517	12.28
XGBoost (GRACE)	Gradient boosting	0.544	11.86

Table 10: Per-group comparison on test records. GRACE improves every history stratum.

Prior tests	n	GRACE R^2	Baseline R^2	GRACE MAE	Base MAE
1–4	69,509	0.522	0.369	12.18	14.17
5–10	99,962	0.555	0.423	11.55	13.35
11–20	131,375	0.571	0.440	11.47	13.40
>20	121,948	0.584	0.445	11.22	13.31

Behavioural Regularities in the Practice-Test Corpus

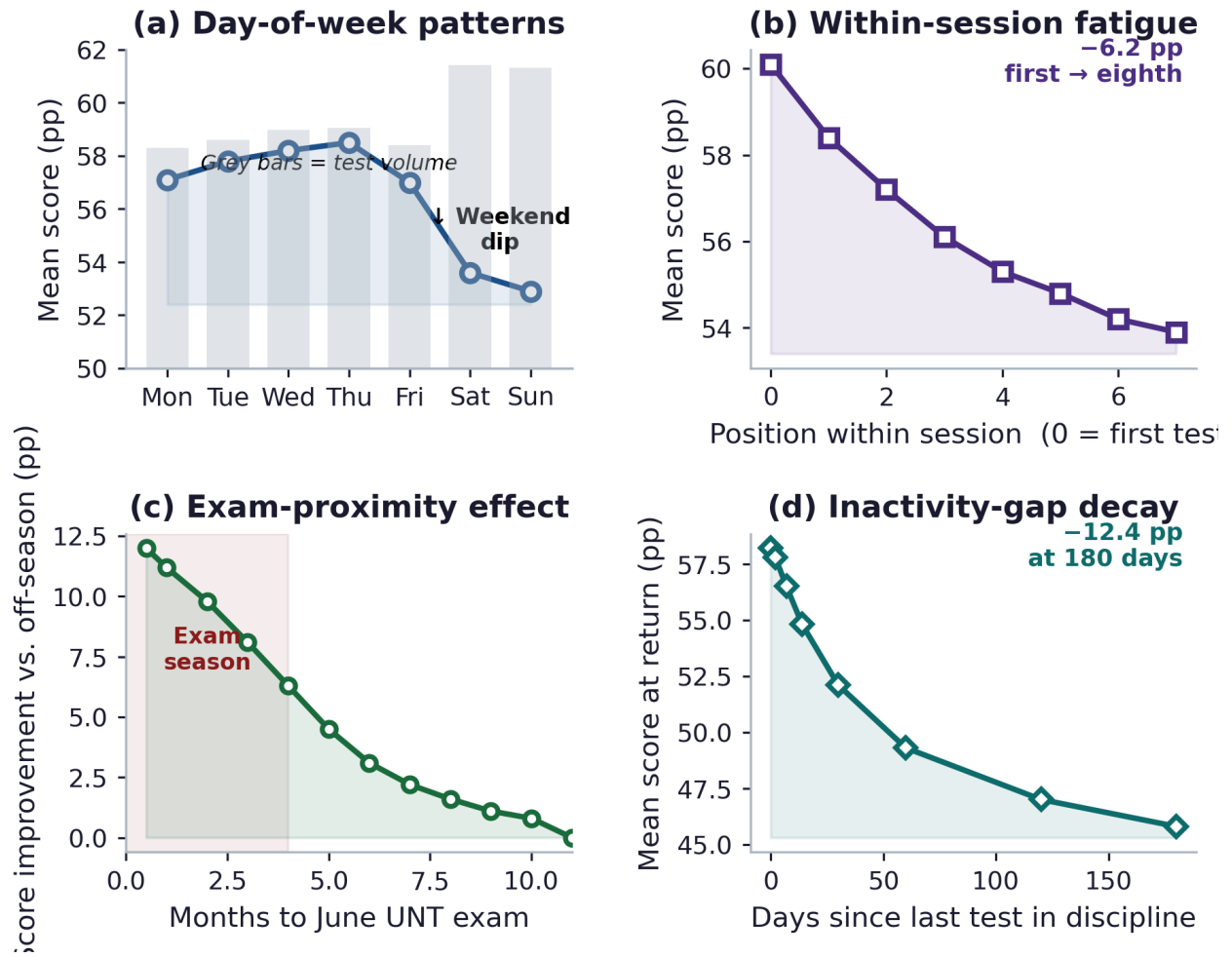


Figure 7: Behavioral regularities. (a) Day-of-week score patterns: mean scores dip on weekends relative to weekdays. (b) Within-session fatigue: mean score drops 6.2 pp from the first to the eighth consecutive test in a session. (c) Exam-proximity effect: score improvement versus off-season peaks close to the June UNT date. (d) Inactivity-gap decay: mean score at return falls 12.4 pp after 180 days of absence.

Table 11: Rolling-origin temporal validation with expanding training windows. GRACE uses 78 features with fold-local static content norms; XGBoost only (fixed params) per fold; ensemble ($\alpha_{\text{XGB}} = 0.15$) applied to the headline result only. [†]Aggregate baseline (9 features; first-attempt rows dropped). [‡]V2 baseline (46 features, XGBoost fixed params, same fold as GRACE; no Optuna).

Fold	Train n	Test n	GRACE R^2	Agg. Base [†]	V2 Base [‡]
2018–20→21	269,172	132,887	0.550	0.295	0.545
2018–21→22	388,771	277,933	0.442	0.048	0.437
2018–22→23	638,910	419,808	0.461	0.273	0.458
2018–23→24	1,016,738	638,492	0.562	0.384	0.570
2018–24→25	1,591,380	440,112	0.554	0.441	0.542
Mean $\pm$ SD	—	—	0.514 $\pm$ 0.051	0.288 $\pm$ 0.151	0.511 $\pm$ 0.053

Table 12: Rating-layer ablation (80/20 chronological split).

Component removed	R^2	ΔR^2 vs full	MAE (pp)
— none: full model (reference)	0.519	—	12.24
— rating-deviation features	0.519	+0.001	12.23
— skill-level ability	0.519	+0.000	12.24
— surprise history	0.515	−0.004	12.32
— inactivity gap	0.510	−0.008	12.36
— aggregate supplement	0.514	−0.005	12.33
— RD time decay ($c = 0$)	0.513	−0.006	12.36

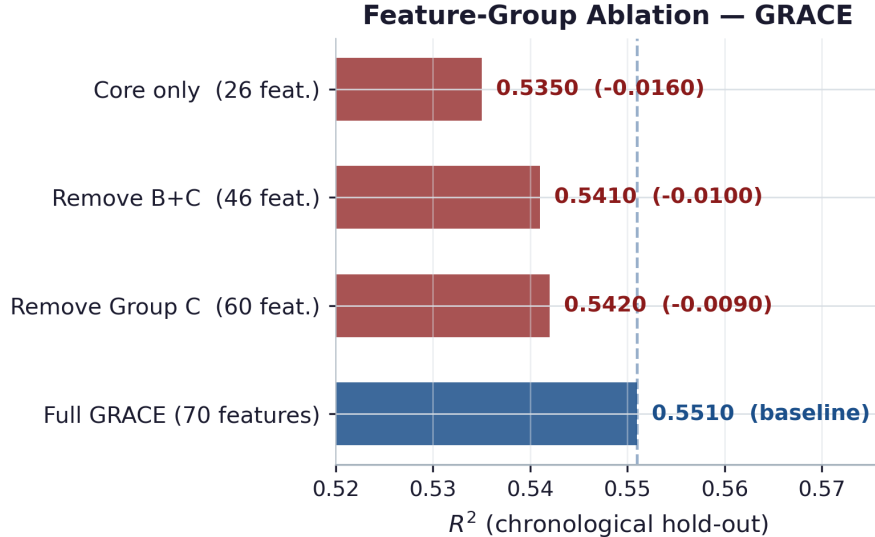


Figure 8: Change in R^2 when individual rating-layer components are removed (80/20 chronological split; reference model $R^2 = 0.519$). Green bars indicate removal has negligible or slightly positive effect; red bars indicate degradation. The inactivity gap and RD time-decay produce the largest individual drops.

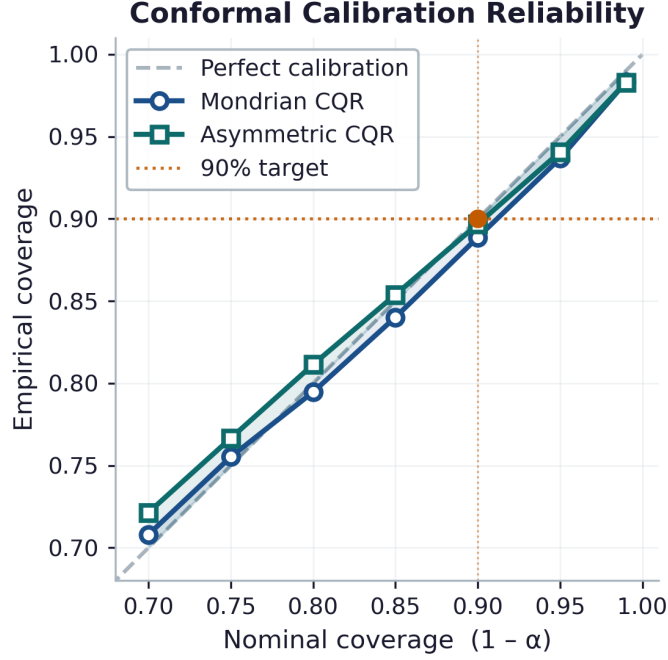


Figure 9: Reliability diagram: nominal vs. empirical coverage across α levels. Asymmetric CQR tracks the ideal diagonal (dashed) closely across the full range; empirical coverage at the 90% nominal target is 0.910.

Table 13: Score-range coverage and mean width by history depth (90% target). Wilson 95% CIs on coverage. Asymmetric CQR with 3rd/97th-percentile quantile heads.

History band	n	Coverage	95% CI	Mean width (pp)
1–4 prior tests	88,378	0.902	[0.900, 0.904]	55.6
5–10 prior tests	99,962	0.915	[0.913, 0.917]	52.0
11–20 prior tests	131,375	0.913	[0.912, 0.915]	51.3
>20 prior tests	121,948	0.908	[0.906, 0.910]	49.2
All attempts	441,663	0.910	[0.909, 0.911]	51.7

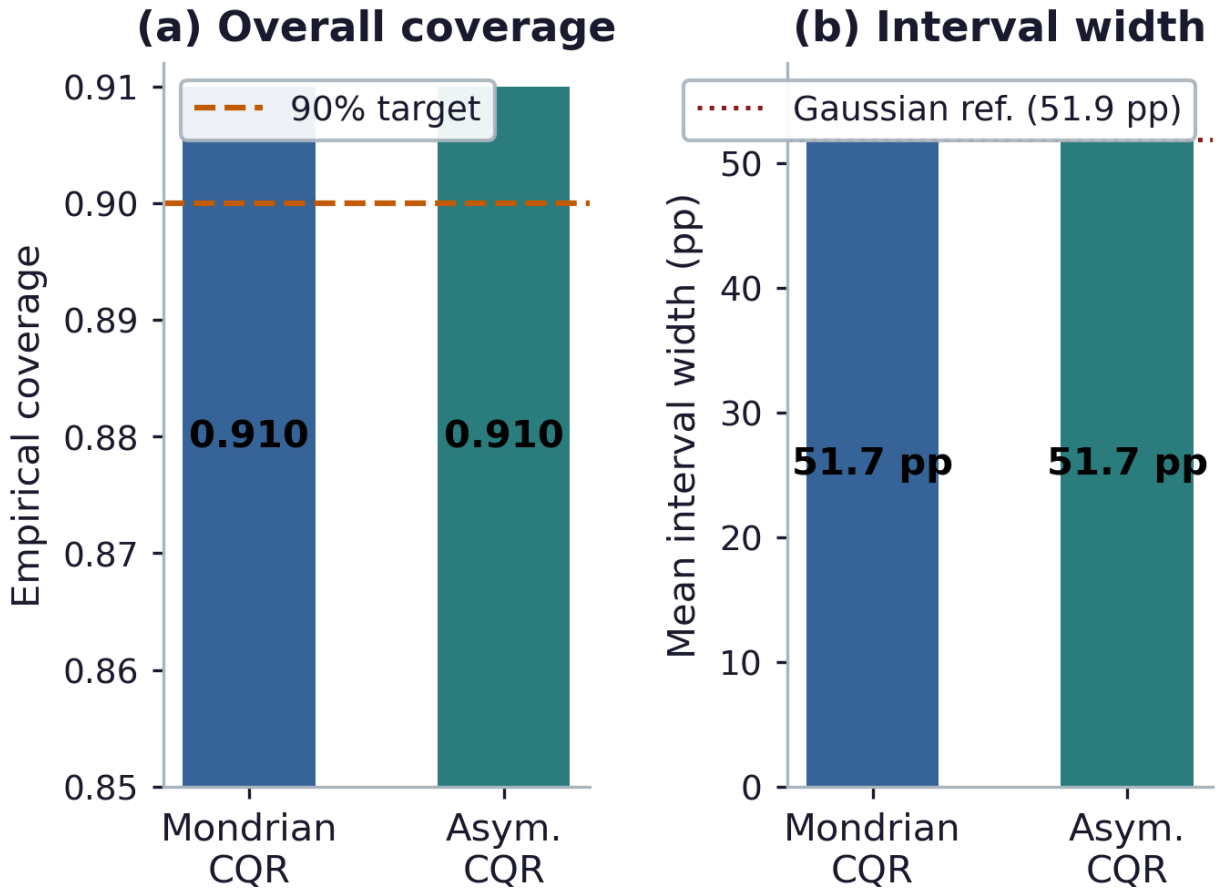


Figure 10: Overall conformal coverage and mean interval width: Mondrian CQR versus Asymmetric CQR. Left: both methods achieve 0.910 empirical coverage, exceeding the 90% nominal target (dashed line). Right: mean interval width; the dotted reference marks the symmetric-Gaussian baseline (51.9 pp); both methods produce identical overall width of 51.7 pp.

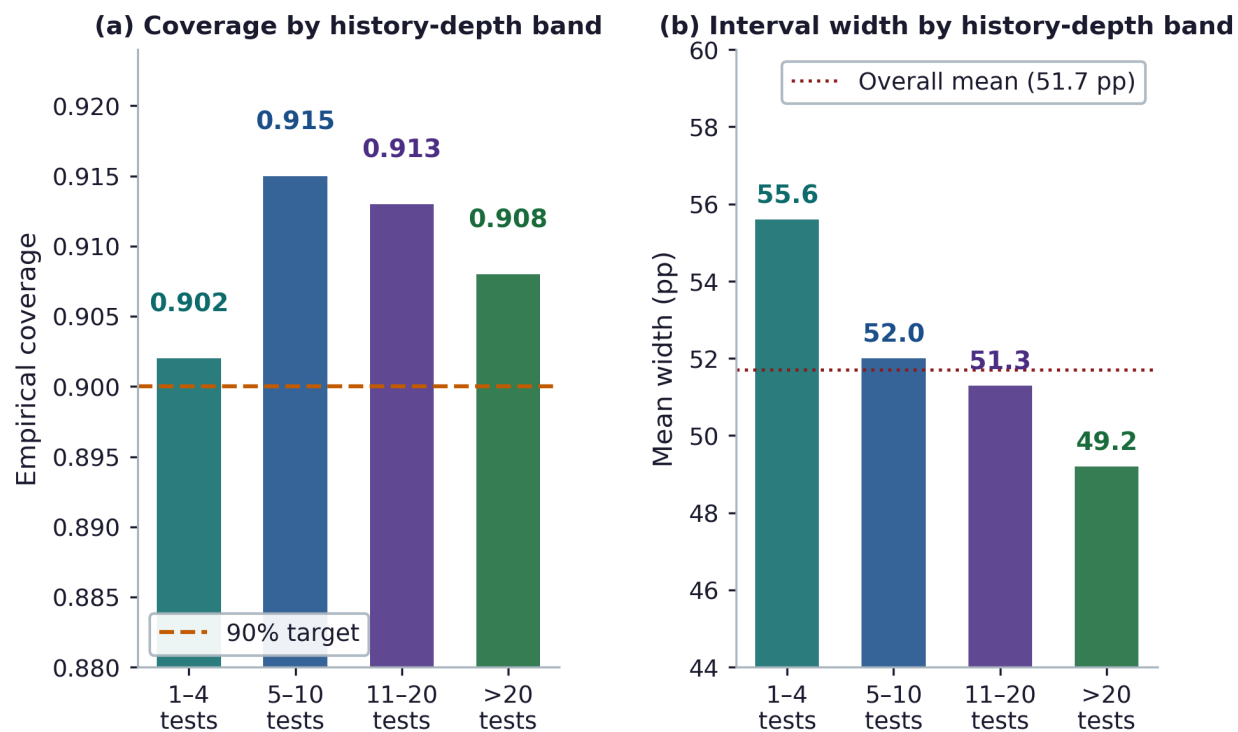


Figure 11: Conformal score-range validity by history-depth band. (a) Empirical coverage per band; all four bands exceed the 90% nominal target (dashed line). (b) Mean interval width per band; the horizontal reference line marks the overall mean of 51.7 pp; cold-start students (1–4 tests) receive wider intervals (55.6 pp) than long-history students (>20 tests: 49.2 pp), reflecting genuine uncertainty reduction with repeated assessment.

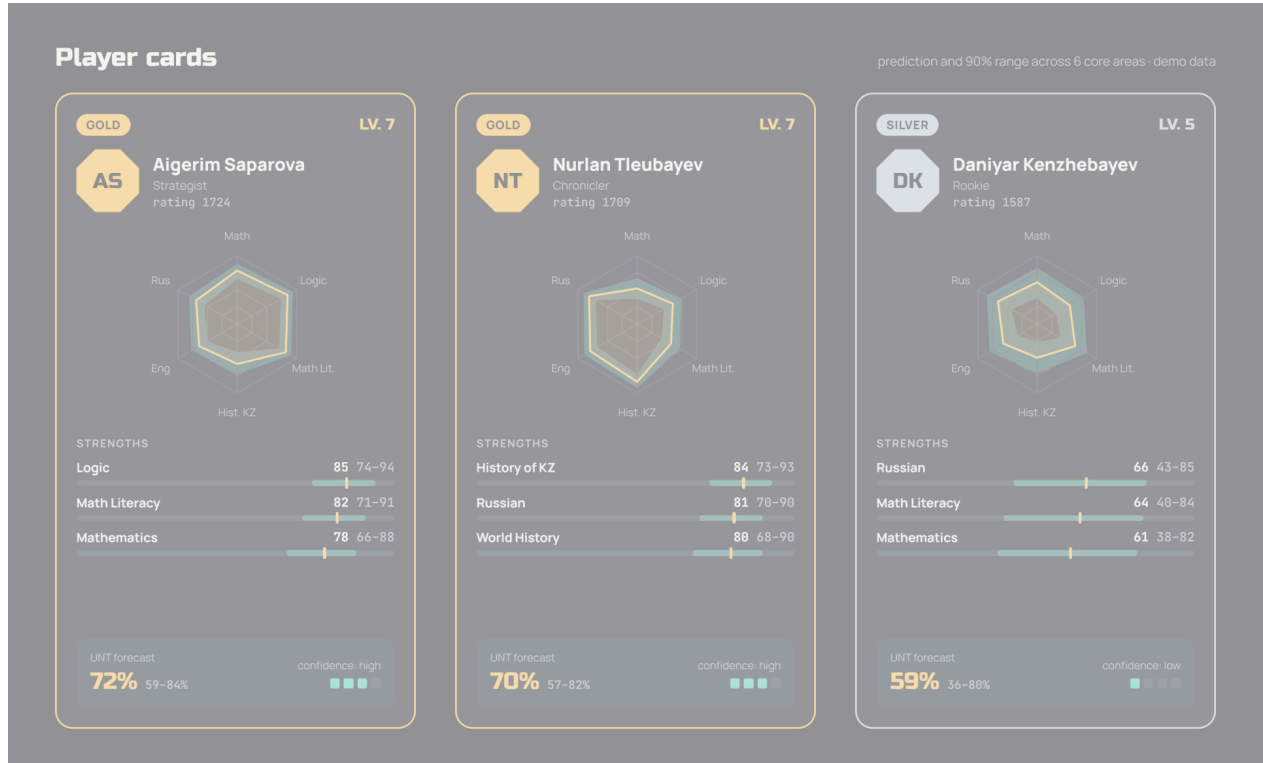


Figure 12: GRACE output as student player cards (demo data; names are illustrative samples not associated with any real individuals). Each card shows the predicted UNT score with 90% score range, per-discipline ability on a radar chart, and a confidence indicator derived from the Glicko rating deviation. Cold-start students (e.g., Daniyar K., Silver tier) receive visibly wider ranges (59–80%) than long-history students (Aigerim S., 59–84%), directly communicating prediction uncertainty.

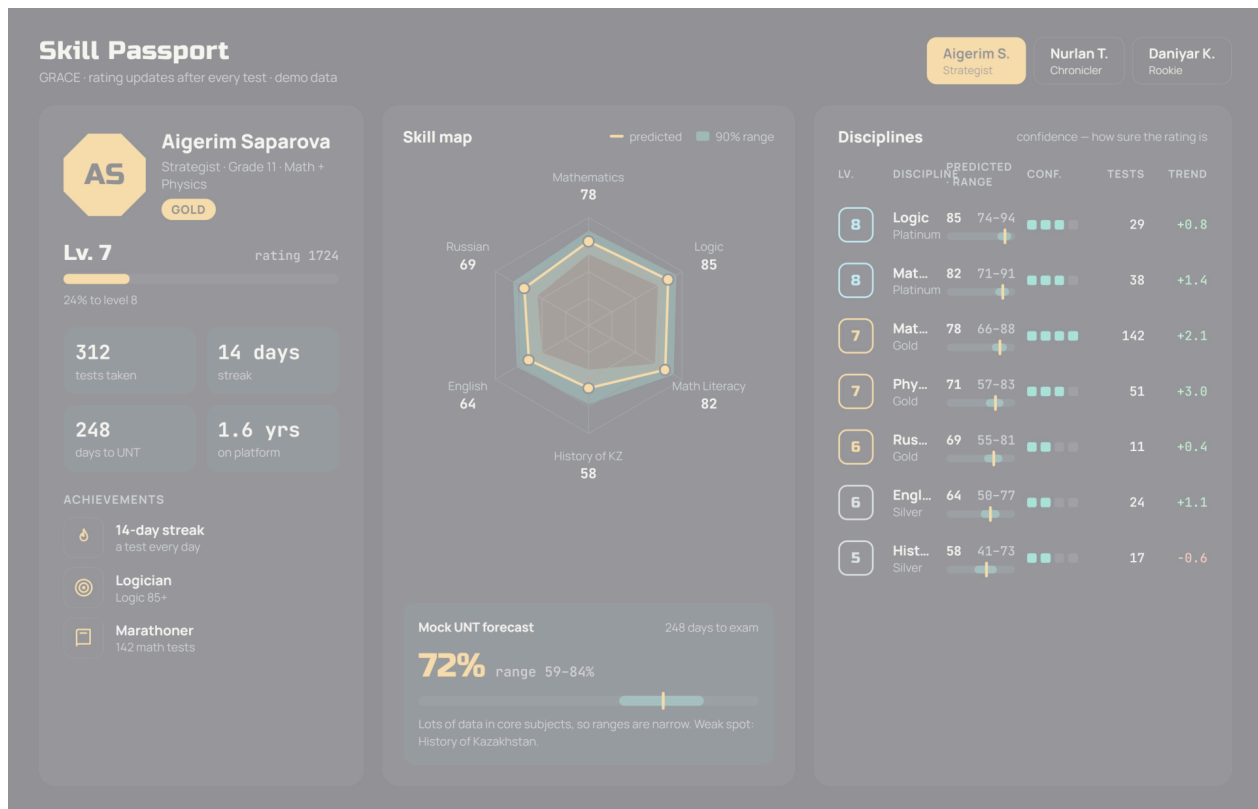


Figure 13: GRACE output as an interactive skill passport (demo data; names are illustrative samples not associated with any real individuals). The dashboard shows per-discipline predicted scores with 90% confidence ranges, Glicko rating level and trend, engagement statistics (tests taken, streak, days to UNT), and a UNT composite forecast with uncertainty band (72%, range 59–84%). Discipline-level confidence bars translate rating deviation directly into a signal educators can act on without statistical training.

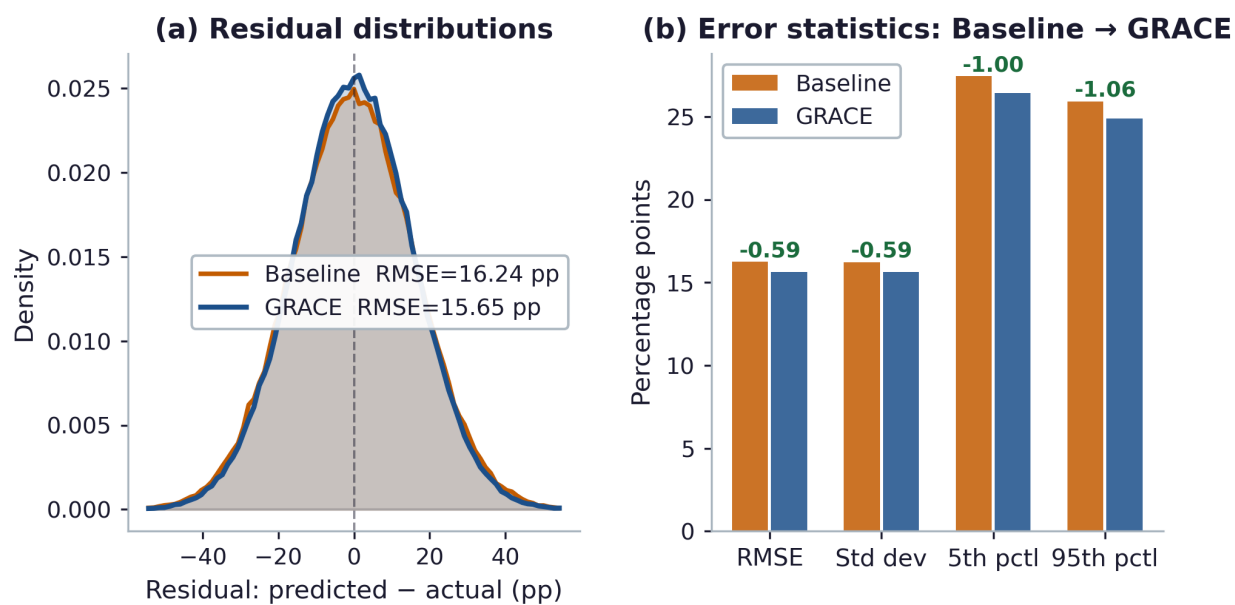


Figure 14: GRACE residual distribution. Left: residual density; the upper tail exceeds the lower, indicating over-prediction of struggling students as the primary error mode. Right: mean bias (predicted - actual) by true-score decile; near-ideal aggregate calibration with modest regression-to-mean tendency at extremes.

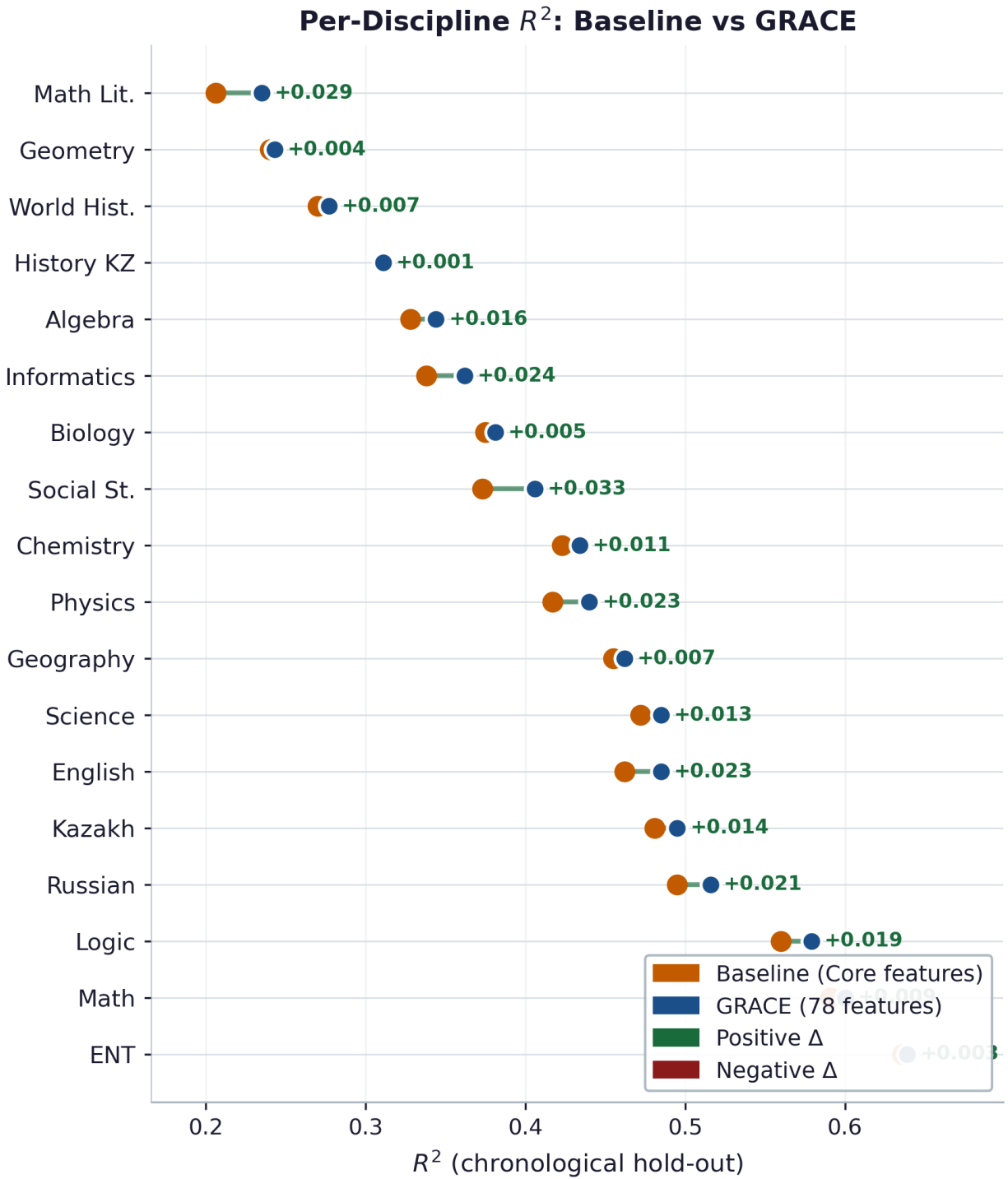


Figure 15: Per-discipline R^2 for GRACE. Disciplines sorted by performance; all disciplines with $n \geq 200$ test records shown. Gains over the aggregate baseline are largest where per-skill difficulty trends are most informative (Mathematics, Logic).